CY1010 Environmental Chemistry

Exam 2 Summary: Segments 3 and 4

Dr. Kishore Natte (organic chemicals, surfactants, polymers, plastics) and Dr. M. Annadhasan (CO, NO_x, VOCs, SO_2, smog, ozone)

Exam 2: 10/10/26 · 50 marks · MCQ · 90 min. Compiled from all four slide decks (Annadhasan L1-L2, L3-L5; Kishore Class 1-3, Class 4) and the 2024 Exam 2 paper. Original slides are embedded after each topic so you can see the diagrams, structures and tables as presented.

★ = asked in 2024 Exam 2 ⚠ = slide inconsistency or likely trap

Contents

- 1. Exam analysis and strategy
- Part A: Dr. Annadhasan — A1 Foundations and context · A2 Measurement and Beer-Lambert · A3 Pollutants overview · A4 Carbon monoxide · A5 Nitrogen oxides · A6 VOCs, gasoline and engines · A7 Catalytic converter · A8 Sulfur dioxide and acid rain · A9 Smog · A10 Stratospheric ozone
- Part B: Dr. Kishore — B1 Framing, economy models, waste · B2 POPs, pesticides, SOCs · B3 Fate, K_{ow}, PBT, TSCA · B4 Soaps and surfactants · B5 Polymers · B6 Plastic waste
- Part C: Number sheet, trap list, quick-fire drill

1. Exam analysis and strategy

What the 2024 Exam 2 looked like (50 MCQ, 90 min)

About 1.8 minutes per question. All questions are one-line recall or small-reasoning items; there is no long calculation. Roughly:

Source (2024 numbering)	Approx. questions	What was tested
Annadhasan (Q2-13, 15-20, 22-23)	~20 (40%)	Ozone instrument, OH radical, five spheres, Rh in converter, O_3 absorption max, Beer-Lambert long path, CO MO nodes/bond order, bridging vs terminal CO, lowest ν(CO), oxy-Hb coordination, CO affinity, Haber pressure, colored NO_x, OH count in smog, dithionic acid, PAN, Dobson unit
Kishore (Q21, 25, 40, 41, 50)	~5-6 (12%)	Toxaphene use and formula, slowest-decomposing material, PCB expansion, TCE name
Other segments (Abhijit/Anup)	~24	Enzymes, Lineweaver-Burk, Krebs cycle, E-factor, atom economy, ionic liquids, Kalundborg, asthenosphere, Cr(VI)

What changes this year

- Exam 2 now covers **only segments 3 and 4** (per syllabus: Kishore + Annadhasan). So about 50 questions come from these ~225 slides, instead of ~25 in 2024.
- Expect Kishore's share to grow a lot, from ~5 to perhaps 20-25 questions. His decks are long (92 + 32 slides) and fact-dense: POP list and uses, dates, K_{ow} ordering, soaps/detergents, polymer synthesis, plastic statistics. Treat names, numbers and definitions as testable.
- Annadhasan's questions were conceptual with a few exact numbers (2143 cm^{-1}, 200 atm, 14.7:1, 2.7×10^{16}). The same decks are used, so the 2024 items are directly reusable as a template.
- Several 2024 answers are visible only in **image slides** (five spheres, oxoacids of sulfur, MO diagram, K_{ow} table, dirty-dozen table). Those images are embedded in this PDF.

Question types to expect

Type	Examples	How to prepare
Direct recall	Haber pressure, DU value, toxaphene use, PCB expansion, Stockholm year	Number sheet in Part C
Concept ordering	Which carbonyl has lowest ν(CO)? Which has highest K_{ow}? Slowest decomposition?	Learn the ν(CO) series and log K_{ow} list
Mechanism or role	Role of Rh vs Pt/Pd; role of OH; why CO binds Hb weakly in the protein	Converter chambers; OH cycle; steric clash slide
Name/formula	$H_2S_2O_6$, $C_{10}H_{10}Cl_8$, PAN, TCE, TCDD	Oxoacid table; POP structures
Trap/discrepancy	Slides disagree on ocean plastic (8 vs 1-2 Mt) and plastic decomposition time	Trap list in Part C

Time plan for the remaining days

- **Day 1:** Annadhasan Part A (A4 CO, A7 converter, A9 smog, A10 ozone are the richest). Do the quick-fire drill at the end.
- **Day 2:** Kishore Part B: B2 POPs/DDT/PCB/dioxin, B3 K_{ow}, B4 surfactants, B5 polymers. Memorise tables.
- **Day 3:** B6 plastic statistics, Part C trap list, then redo the 2024 paper timed (the Annadhasan and Kishore questions only), and re-read ⚠ items.

Part A: Dr. Annadhasan — Atmospheric pollution and ozone

A1. Foundations and context (L1-L2)

Photosynthesis and the nitrogen cycle

- Photosynthesis: $H_2O + CO_2 \rightarrow 1/n\ (CH_2O)_n + O_2$, $\Delta H = +470$ kJ/mol (uphill). Two Nobel Prizes quoted (e.g. 1992 Rudolph Marcus, electron-transfer theory, bacterial photosynthesis).
- **Haber-Bosch** ★: $N_2 + 3H_2 \rightarrow 2NH_3$, Fe/Mo catalyst, **200 atm**, **450 °C**; ammonia → urea. Uses ~2% of global energy; steam methane reforming supplies >80% of that energy; ~500 Mt CO_2 (~1.8% of global CO_2). Nobel Prizes: Haber 1918, Bosch 1931. Ammonia → nitric acid → nitrates (explosives).
- Biological fixation: leguminous plants (*Pisum sativum*), root nodules with N_2-fixing bacteria. **Nitrogenase**: MoFe protein is an $\alpha_2\beta_2$ heterotetramer with P-clusters and FeMo-cofactor; the Fe protein donates electrons; $N_2 \rightarrow NH_3$.

Nitric oxide (NO) in biology (slides 9-10)

- NO is a signalling molecule: dilates blood vessels, reduces platelet stickiness, prevents plaque formation, reduces LDL oxidation, restricts superoxide release, inhibits smooth-muscle proliferation.
- **1998 Nobel Prize** in Physiology/Medicine: Furchgott, Ignarro, Murad, for NO as a signalling molecule in the cardiovascular system. Pathways: L-arginine + O_2 → NO via NO synthase; and diet nitrate → nitrite → NO. NO ends as NO_3^-/NO_2^-. We lose ~85% of NO-making ability with age.

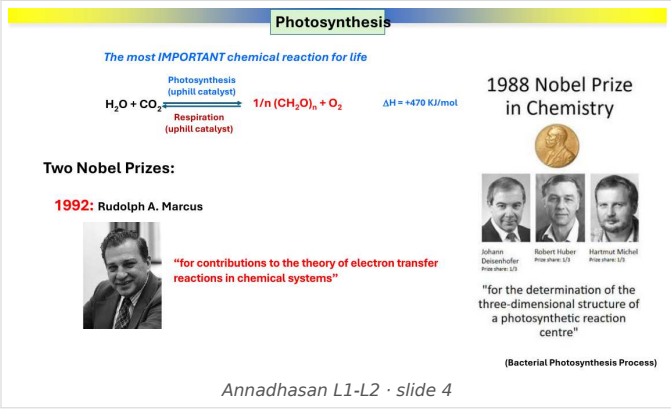
Five spheres and biogeochemical cycles (slides 11-12) ★

- Five spheres: **atmosphere, hydrosphere, geosphere, biosphere, anthrosphere** (the 2024 Q4 asked which is NOT one). Matter and energy are exchanged largely through biogeochemical cycles; the carbon cycle is key to climate change (combustion, respiration, photosynthesis, dissolution in oceans, sediments, fossil carbon).

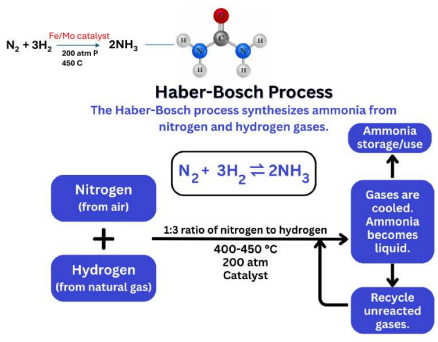
Environmental history and pollutant classes (slides 13-18, 23)

- Past practices (1954 industry quote) produced polluted air and water and hazardous waste sites. From ~1970 laws relied on **end-of-pipe controls** (generate then remove); costly but effective at limiting releases.
- Industrial pollutants: **organic** vs **inorganic**. Inorganic → heavy-metal ions (Cd, Pb, Cu; venomous metals Hg, Cr, Pb, Ni, Cd; precious metals Pd, Pt, Ag, Au; radionuclides U, Cs, Rb) and non-metal ions (phosphates, nitrates, sulfides).
- Organic pollutants: **persistent organic pollutants** (industrial: OCPs, PCBs, PBDEs; unintentional: PAHs, dioxins) and other organic (emerging) pollutants: steroid hormones, endocrine disruptors, pharmaceuticals and personal-care products, surfactants such as LABs, other brominated flame retardants.
- **Persistent pollutants** (slide 18): resist breakdown; long-lasting (sunlight, water, microbes do not remove them); bioaccumulative (fatty tissue; biomagnification up the food chain); mobile (wind and water carry them across borders).
- Exposure pathways: oral ingestion, inhalation, dermal contact. Sources: shipping, aviation, industry, residential heating, biomass burning, biogenic, road traffic.

Slides from the course deck (Annadhasan L1-L2)



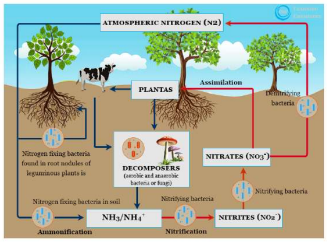
Annadhasan L1-L2 · slide 4



Annadhasan L1-L2 · slide 5

Nitrogen Fixation

Nitrogen-fixing Leguminous Plant: *Pisum sativum*



Highly energy-intensive process consuming around 2% of global energy output each year (steam methane reforming accounts for over 80% of the energy required) and producing as a result about 500 million tonnes of carbon dioxide (about 1.8% of global carbon dioxide emissions)

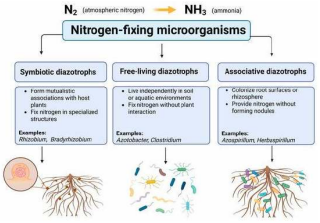
Haber and **Bosch** were later **awarded Nobel Prizes**, in **1918** and **1931** respectively, for their work in overcoming the chemical and engineering problems of large-scale, continuous-flow, high-pressure technology.

Synthetic ammonia from the Haber process was used for the production of nitric acid, a precursor to the nitrates used in explosives.

Nitric Oxide (NO)?

Annadhasan L1-L2 · slide 6

The roots of leguminous plants have nodules that contain N_2-fixing bacteria

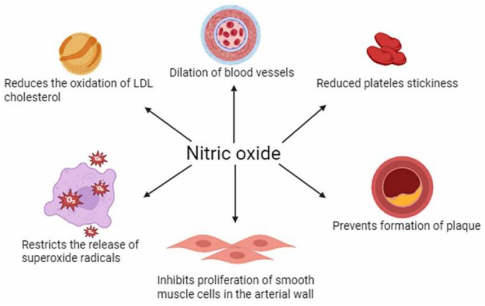




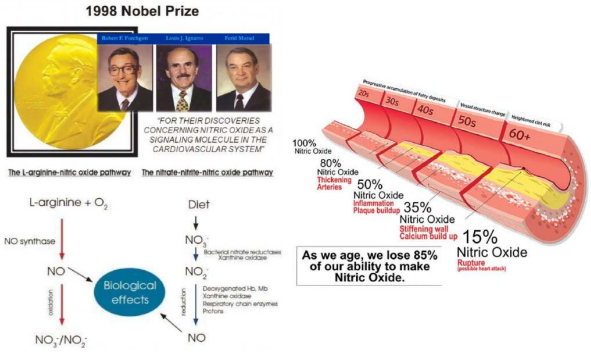
Nitrogenase: key enzyme for biological nitrogen fixation.

The molybdenum-iron (MoFe) protein is the large, catalytic component (component 1) of the nitrogenase enzyme complex. It is a $\alpha_2\beta_2$ heterotetramer that binds metal cofactors (P-clusters and FeMo-cofactors), receiving electrons from the Fe protein to reduce atmospheric nitrogen (N_2) into usable ammonia (NH_3).

Annadhasan L1-L2 · slide 7

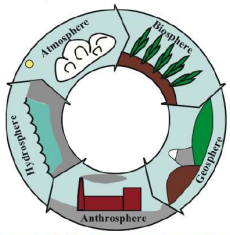


Annadhasan L1-L2 · slide 9



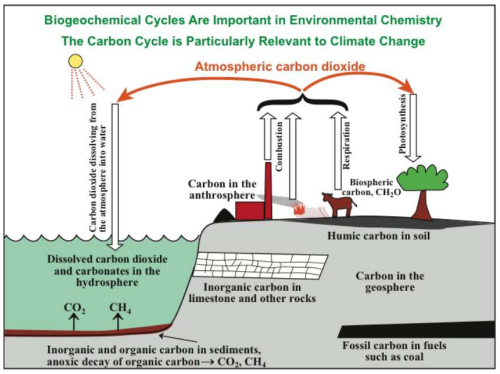
Annadhasan L1-L2 · slide 10

The Five Spheres of the Environment

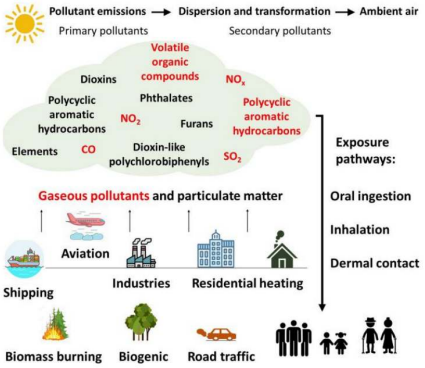


Matter and Energy are Exchanged Largely through Biogeochemical Cycles

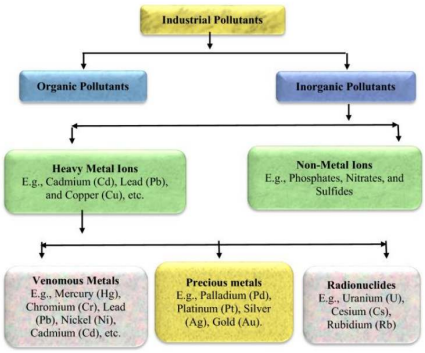
Annadhasan L1-L2 · slide 11



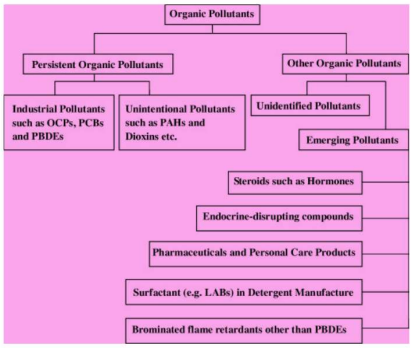
Annadhasan L1-L2 · slide 12



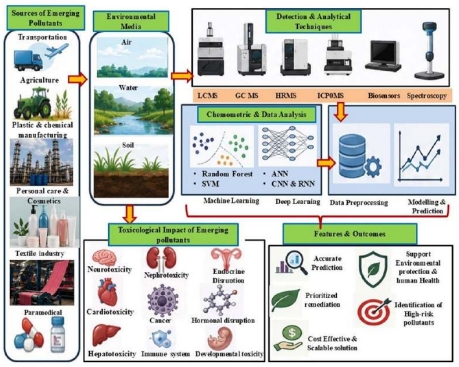
Annadhasan L1-L2 · slide 15



Annadhasan L1-L2 · slide 16



Annadhasan L1-L2 · slide 17

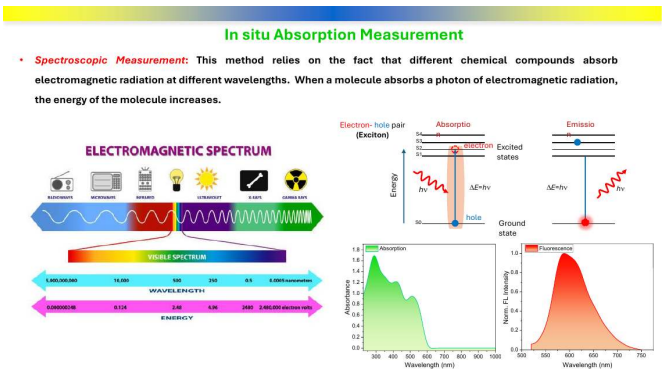


Annadhasan L1-L2 · slide 23

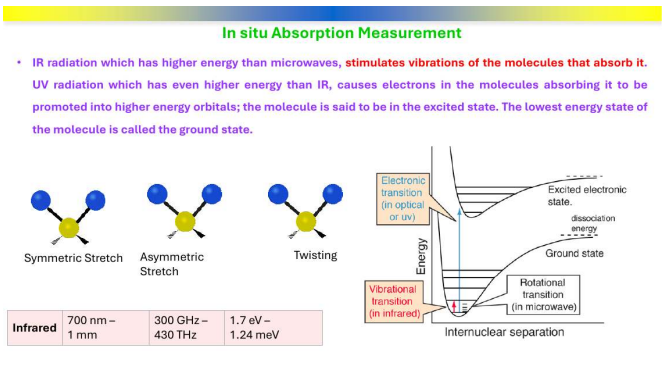
A2. Measurement of air pollutants and Beer-Lambert (L1-L2)

- Units: μg/m^3, ppm, ppb; particulate size in μm. N_2 is >78% of air while reactive species such as ozone are at ppm level, so concentration spans ~10^4.
- Three approaches: **in situ** (sample placed inside the instrument), **remote sensing** (beam from satellite, aircraft, space shuttle or ground through the atmosphere), **ex situ** (collect and analyse in a lab). Principles are the same. Car and smokestack emissions are hard to measure because exhaust is hot and fast.
- Spectroscopic absorption: molecules absorb at characteristic wavelengths; absorbing a photon raises molecular energy. **Microwave → rotation; IR → vibration; UV/visible → electronic excitation; X-ray/γ → bond breaking and ionisation.** Ground state is the lowest energy state.
- Irradiance P (energy per second per unit area); a monochromator selects one wavelength; P_0 in, P out.
- **Transmittance T = P/P_0** (0 to 1; %T = 100T). **Absorbance A = log(P_0/P) = −log T**; no absorption → P = P_0, A = 0. A is dimensionless.
- **Beer-Lambert law: A = ε·b·C**, ε = molar absorptivity (extinction coefficient; characteristic of the molecule at a wavelength), b = path length, C = concentration.
- **Ozone measurement ★**: UV absorption by a UV photometer. O_3 absorbs strongly in the UV (maximum 250-260 nm ★) while O_2, N_2, H_2O do not absorb and do not interfere. Ozone is present at ppm-level or less, so the cell is made **very long** (A ∝ path length) ★.

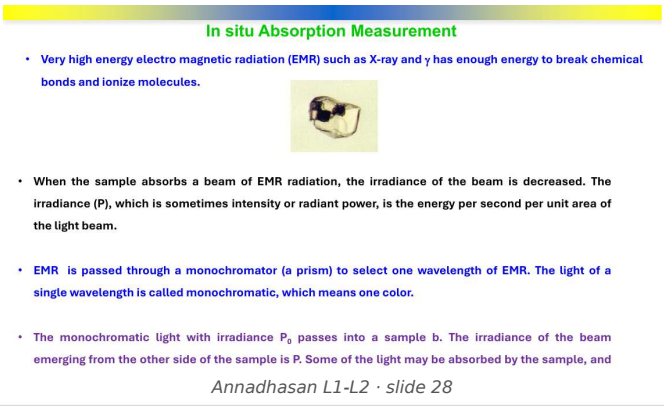
Slides from the course deck (Annadhasan L1-L2)



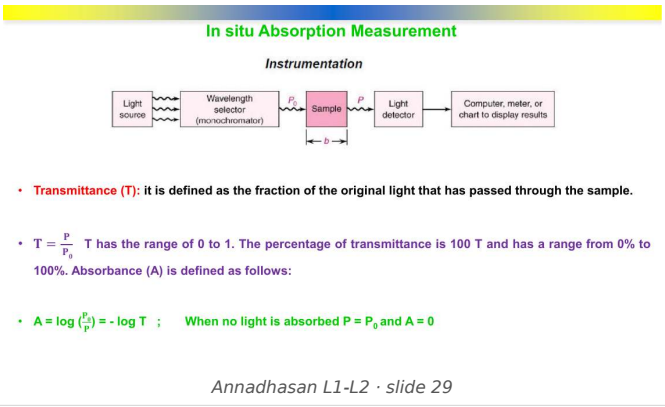
Annadhasan L1-L2 · slide 24



Annadhasan L1-L2 · slide 27



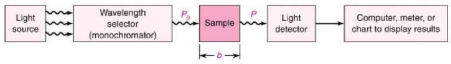
Annadhasan L1-L2 · slide 28



Annadhasan L1-L2 · slide 29

In situ Absorption Measurement

Instrumentation

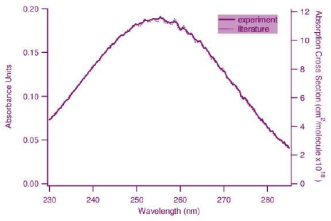


- Absorbance is important because it is directly proportional to the concentration (C) of the absorbing molecules in the sample. A is dimensionless.
- The Beer-Lambert or simply Beer's Law: $A = \varepsilon C l$
- A = ε. b. C (b = path length cm or m etc., C = concentration moles/L, ppm, ppb etc.,)
- ε = molar absorptivity or extinction coefficient. It is the characteristics of the molecule that indicates how much light it will absorb at a particular wavelength.

Annadhasan L1-L2 · slide 30

In situ Ozone measurement

- *Ultraviolet* absorption has been used to measure the concentration of ozone in stratospheric air. The figure shows that ozone has a strong absorption in the UV region. Other atmospheric gases such as O_2, N_2 and H_2O do not absorb UV radiation and thus do not have to be removed from the air sample as they do not interfere with the ozone measurement.

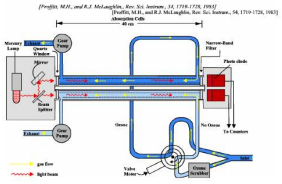


Annadhasan L1-L2 · slide 31

In situ Ozone measurement

- The concentration of ozone in the stratosphere is less than 100 ppm, so the spectrophotometer that is used to make the measurement must be capable of measuring a small absorbance. This is most easily done by making the sample cell very long. Because Beer's law says that absorbance is proportional to the sample path length and because we can get a lot of samples easily, a long sample path length will greatly improve our ability to measure the ppm level of Ozone.

Ozone:Dual Beam UV-Photometer



Annadhasan L1-L2 · slide 32

A3. Pollutants overview (L1-L2)

- **Primary pollutants** are emitted directly (CO from cars, SO_2 from coal and factories); **secondary** form by reactions in the air (H_2SO_4, HNO_3, SO_4^{2-}, NO_3^- for acid rain; O_3 and photochemical oxidants for photochemical smog; PAN).
- An air pollutant is a substance present at a concentration sufficient to harm humans, animals, vegetation or materials. About **90%** of air pollution comes from primary pollutants: CO, SO_2, NO_x, VOCs (mostly hydrocarbons), suspended particles.

Pollutant	Abbrev.	Type (slide 33)
Carbon monoxide	CO	Primary
Sulfur dioxide	SO_2	Primary
Ozone	O_3	Secondary
Nitrogen dioxide	NO_2	Secondary
Hydrocarbons (VOCs)	HC	Primary and secondary
Particulate matter	PM	Primary and secondary
Lead	Pb	Primary and secondary

Slides from the course deck (Annadhasan L1-L2)

Basic Pollutants

Basic Pollutants

Pollutant	Abbreviation	Type
Carbon Monoxide	CO	Primary
Sulfur Dioxide	SO_2	Primary
Ozone	O_3	Secondary
Nitrogen Dioxide	NO_2	Secondary
Hydrocarbon Compounds (also called VOCs – volatile organic compounds)	HC	Primary & Secondary
Particulate Matter	PM	Primary & Secondary
Lead	Pb	Primary & Secondary

Annadhasan L1-L2 · slide 33

A4. Carbon monoxide (L1-L5)

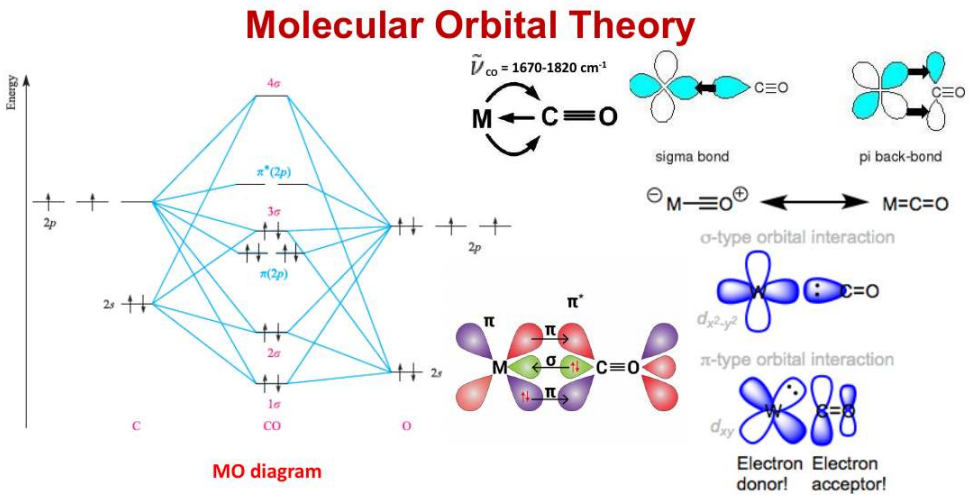
Sources and health

- Incomplete combustion of carbon fuels, mainly gasoline vehicles. Sources: transportation **68.4%**, industry 11.3%, solid waste 8.1%, misc 10.2%, stationary fuel combustion 2%. Natural: CH_4 oxidation ($2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$).
- Complete: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$. Incomplete: $2C_8H_{18} + 17O_2 \rightarrow 16CO + 18H_2O$.
- Poisoning (ppm): **35** headache and dizziness · **200** headache and loss of judgement · 3200 headache, dizziness, nausea · 6400 respiratory arrest, death < 20 min · 12800 unconscious, death < 3 min. Symptom chart: mild (headache, dizziness, nausea, confusion), moderate (weakness, breathlessness, blurred vision), severe (chest pain, loss of consciousness, seizures, death).

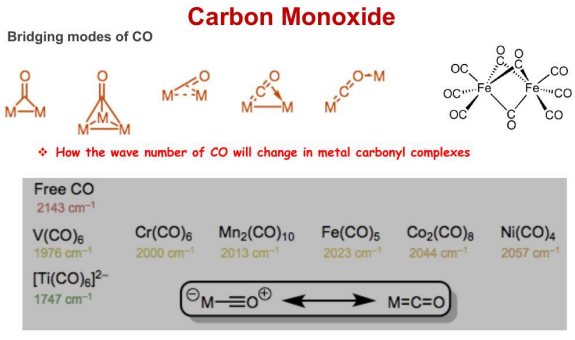
Molecular orbitals and IR of CO ★

- CO is isoelectronic with N_2: **bond order 3**; the σ and π MOs have nodes (the 2024 paper asked for number of nodes in a given MO and bond order). Free CO stretches at **2143 cm^{-1}**; in metal carbonyls ν(CO) falls (~1670-1820 cm^{-1} for bridging) because of π back-donation into CO π* (M=C=O resonance form).
- **Bridging CO has lower ν(CO) than terminal CO ★**. More electron-rich metal or negative charge → more back-donation → lower ν.
  - $V(CO)_6$ 1976 · $Cr(CO)_6$ 2000 · $Mn_2(CO)_{10}$ 2013 · $Fe(CO)_5$ 2023 · $Co_2(CO)_8$ 2044 · $Ni(CO)_4$ 2057 cm^{-1}; $[Ti(CO)_6]^{2-}$ **1747** (the lowest) ★.
- IR spectrum example (Cp-Fe dimer): one band in the bridging region means the two bridging COs are almost co-linear; two terminal bands means the two COs are not co-linear.
- **Mutual exclusion principle**: for centrosymmetric molecules, IR-active modes are Raman-inactive and vice versa. CO_2: symmetric stretch 1285 cm^{-1} (Raman only), asymmetric stretch 2349 (IR), bend 666 (IR, degenerate). IR table slides: C=O 1680-1750, O-H 3230-3550 etc.

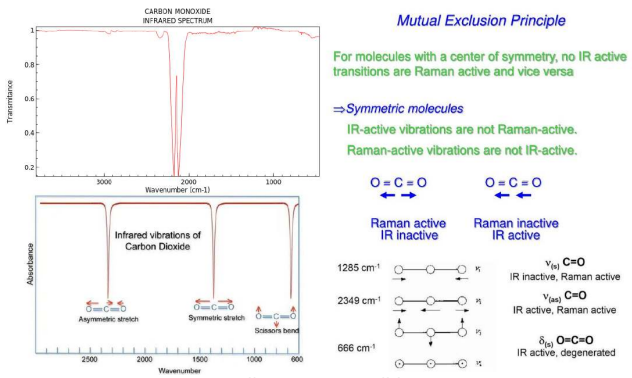
Slides from the course deck (Annadhasan L3-L5)



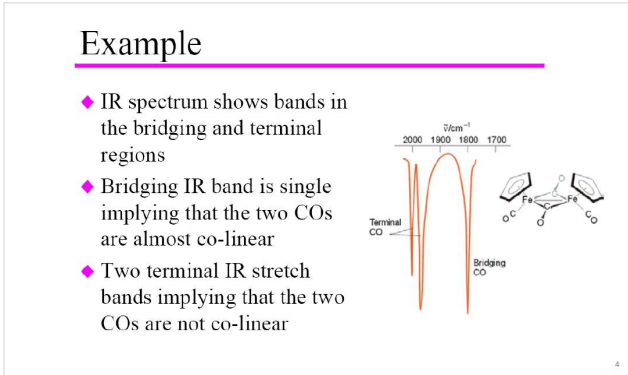
Annadhasan L3-L5 · slide 1

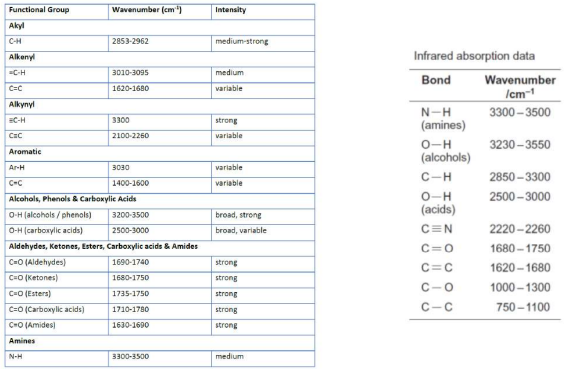


Annadhasan L3-L5 · slide 2



Annadhasan L3-L5 · slide 3



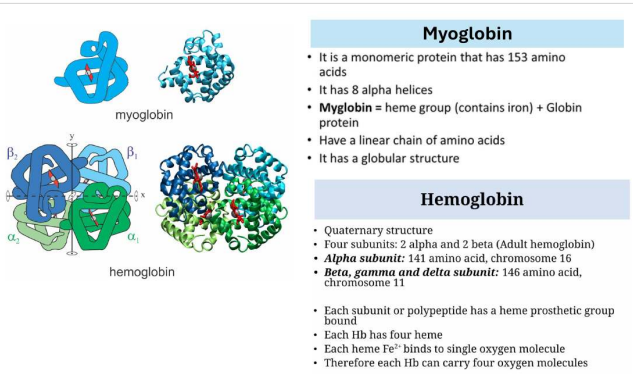


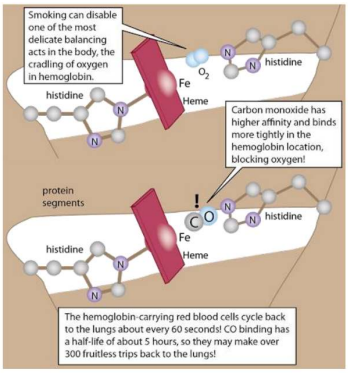
Functional Group	Wavenumber (cm⁻¹)	Intensity
Alkyl		
C-H	2853-2962	medium-strong
Alkenyl		
=C-H	3010-3095	medium
C=C	1620-1680	variable
Alkynyl		
≡C-H	3300	strong
C≡C	2100-2260	variable
Aromatic		
Ar-H	3030	variable
C=C	1400-1600	variable
Alcohols, Phenols & Carboxylic Acids		
O-H (alcohols / phenols)	3200-3500	broad, strong
O-H (carboxylic acids)	2500-3000	broad, variable
Aldehydes, Ketones, Esters, Carboxylic acids & Amides		
C=O (Aldehydes)	1690-1740	strong
C=O (Ketones)	1680-1750	strong
C=O (Esters)	1735-1750	strong
C=O (Carboxylic acids)	1710-1780	strong
C=O (Amides)	1630-1690	strong
Amines		
N-H	3300-3500	medium

Infrared absorption data

Bond	Wavenumber /cm⁻¹
N—H (amines)	3300 – 3500
O—H (alcohols)	3230 – 3550
C—H	2850 – 3300
O—H (acids)	2500 – 3000
C≡N	2220 – 2260
C=O	1680 – 1750
C=C	1620 – 1680
C—O	1000 – 1300
C—C	750 – 1100

Annadhasan L3-L5 · slide 5

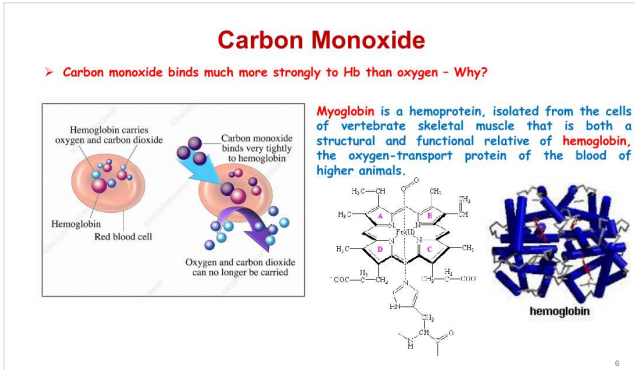


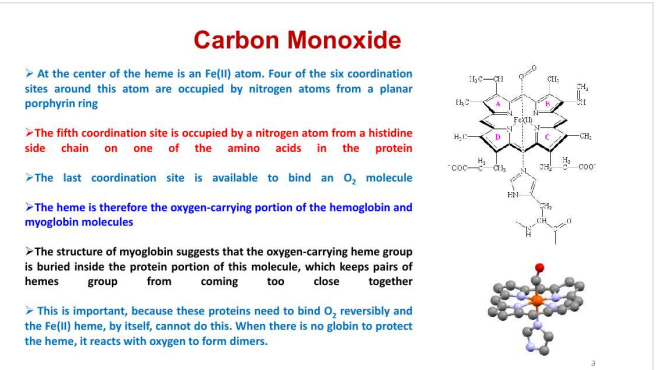


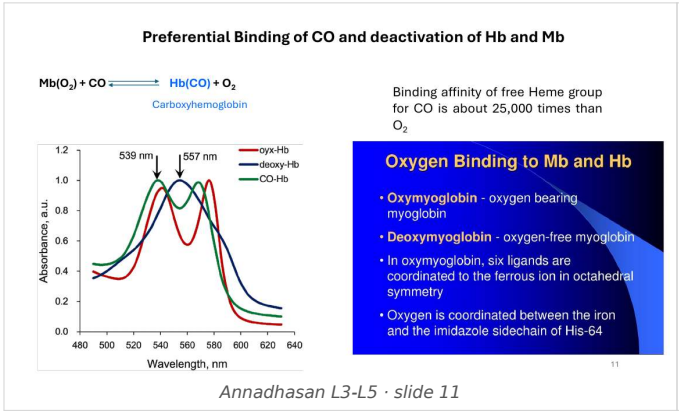
Hemoglobin, myoglobin and CO ★

- Myoglobin (Mb): single polypeptide, ~153 aa (Mr ~16,700-17,500), 8 α-helices, one heme; stores O_2 in muscle and facilitates diffusion; **hyperbolic** saturation curve, higher O_2 affinity.
- Hemoglobin (Hb): quaternary structure, 2α + 2β (α 141 aa, chromosome 16; β 146 aa, chromosome 11), four hemes, carries four O_2; **sigmoidal** (cooperative) curve; lower pH shifts the curve right (Bohr effect).
- **Heme ★**: Fe(II) in a planar porphyrin; **four N from the porphyrin, the fifth site is the proximal histidine N, the sixth binds O_2**. So oxyhemoglobin is **six-coordinate** (2024 asked this). The globin buries the heme so pairs of hemes cannot approach; free Fe(II) heme reacts with O_2 to form dimers and cannot bind O_2 reversibly.
- **CO binding**: $Mb(O_2) + CO \rightarrow Mb(CO) + O_2$; Hb → carboxyhemoglobin (HbCO). Binding affinity of the **free heme group for CO is ~25,000× that for O_2** ★ (slide 11). In Hb the slide on derivatives says ~200×. Reason for the reduction in the protein: distal His causes a **steric clash**, linear Fe-C-O is not possible, Fe^{2+} is low spin, bonding is weak when CO is bound in the angular fashion.
- HbCO half-life is about 5 h; red blood cells cycle to the lungs every ~60 s, so CO-blocked cells make 300+ fruitless trips.
- **Hb derivatives**: oxy-Hb (O_2), deoxy-Hb, carbamino-Hb (CO_2, non-covalent to globin; ~23% of CO_2 transport), met-Hb (Fe^{3+}), carboxy-Hb (CO), sulf-Hb (green blood), glycated Hb (HbA1c, glucose; diabetes marker).

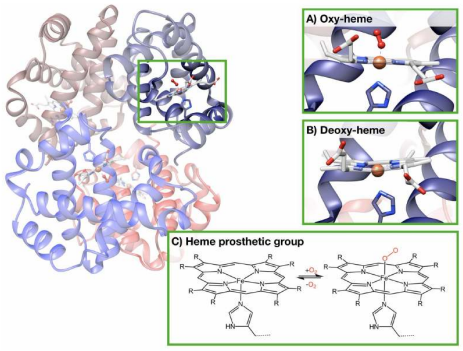
Slides from the course deck (Annadhasan L3-L5)



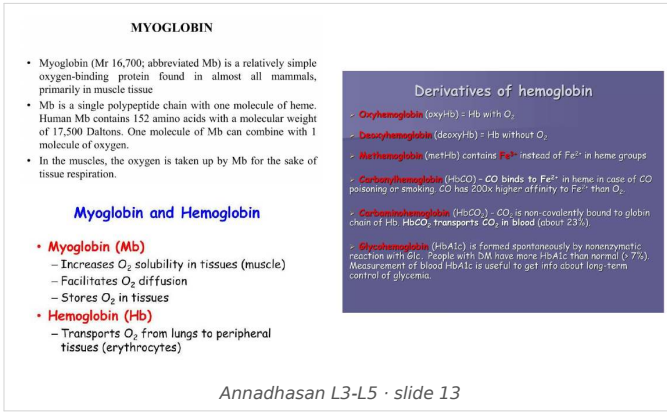




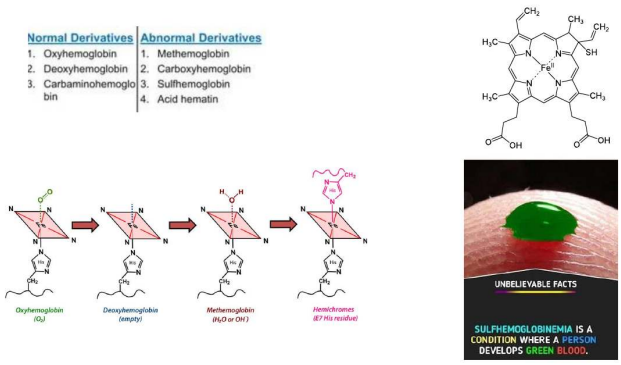
Annadhasan L3-L5 · slide 11



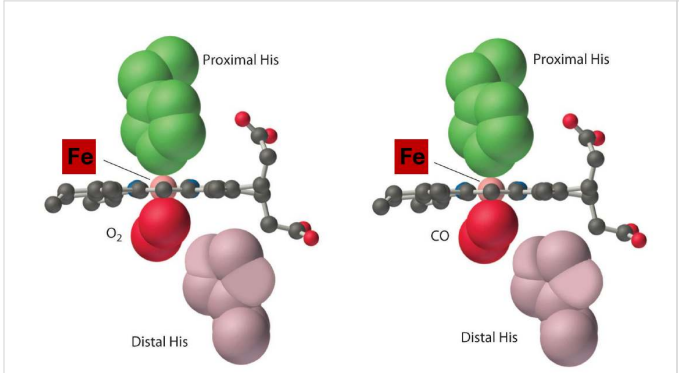
Annadhasan L3-L5 · slide 12



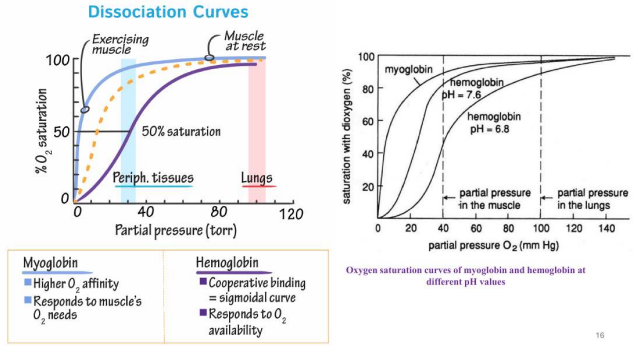
Annadhasan L3-L5 · slide 13



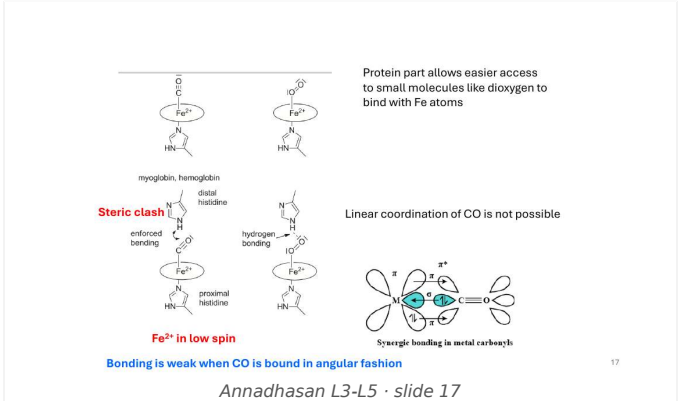
Annadhasan L3-L5 · slide 14



Annadhasan L3-L5 · slide 15



Annadhasan L3-L5 · slide 16



Annadhasan L3-L5 · slide 17

Fate of atmospheric CO ★

- Main sink is oxidation by the **hydroxyl radical**, the **'detergent of the atmosphere'** ★: CO + HO• → CO_2 + H•. ⚠ The slide labels this as stratospheric; in reality it is mainly tropospheric. Residence time of CO: **2 weeks to 3 months**.
- Hydroperoxyl radical: O_2 + H → HOO•; HOO• + NO → HO• + NO_2; 2HOO• → H_2O_2 + O_2; H_2O_2 + hν → 2HO•. OH source: O_3 + hν (<310 nm) → O* + O_2; O* + H_2O → 2OH.
- **Soil** is a major sink (~40% of total CO uptake) through microbes. Soil parameters: water content, temperature, organic matter, pH, soil type. Humic acid average formula $C_{187}H_{186}O_{89}N_9S_1$.

Slides from the course deck (Annadhasan L3-L5)

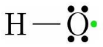
Carbon Monoxide

Fate of Atmospheric CO

Most of the CO molecules are removed from the atmosphere by oxidation (natural) in the stratosphere.

The most important oxidizing species in the air

❑ Hydroxyl radical OH

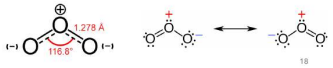


❑ Nitrate radical NO_3



'detergent of the atmosphere'.

❑ Ozone molecule O_3



18

Annadhasan L3-L5 · slide 18

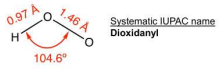
Carbon Monoxide

Fate of Atmospheric CO

It is generally believed that CO is removed from the atmosphere by a reaction with hydroxyl radical.

Oxidized by reaction with hydroxyl radical

• $CO + HO^{\bullet} \rightarrow CO_2 + H$ (stratosphere)



Systematic IUPAC name
Dioxidanyl

- CO is oxidized to CO_2 in the stratosphere, where it migrates *via* convection, turbulence, and mixing
- The reaction is fast and independent of temperature; thanks to it, the residence time of CO in the atmosphere is relatively short, from 2 weeks to 3 months.

19

Annadhasan L3-L5 · slide 19

Carbon Monoxide

Production of hydroperoxyl radical

• $O_2 + H \rightarrow HOO^{\bullet}$

Additional reactions

• $HOO^{\bullet} + NO \rightarrow HO^{\bullet} + NO_2$

• $HOO^{\bullet} + HOO^{\bullet} \rightarrow H_2O_2 + O_2$

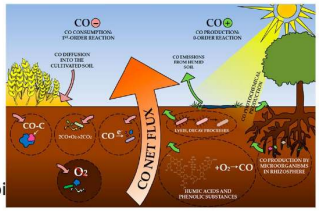
• $H_2O_2 + h\nu \rightarrow HO^{\bullet}$

$O_3 + h\nu$ (sun) → $O + O_2$ (wavelength <310 nm)

$O + O_2 + M \rightarrow O_3 + M$

$O + H_2O \rightarrow$ 2OH 3%

CO Fate in Soi



- Soil is considered one of the main sinks of atmospheric CO, responsible for 40% of total consumption
- Soil is a sink for CO
- Soil microorganisms metabolize CO

Humic acid has the average chemical formula $C_{187}H_{186}O_{89}N_9S_1$

20

Annadhasan L3-L5 · slide 20

Slides from the course deck (Annadhasan L1-L2)

Chemistry of Troposphere

- Carbon Monoxide
  - Source of Carbon monoxide
  - The main anthropogenic source of CO is the combustion of gasoline in automobile engines

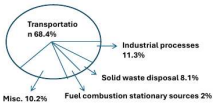
1. Gasoline is a complex mixture of hydrocarbons (HCs)
2. If this is ignited in an adequate supply of oxygen; the products are CO_2 and H_2O

$2\,C_8H_{18} + 25\,O_2 \longrightarrow 18\,H_2O + 16CO_2$

1. In the confined space of the internal combustion of the engine O_2 supply is limited, and combustion is incomplete. Thus, CO is formed and released into the atmosphere in automobile exhaust
2. $2\,C_8H_{18} + 17\,O_2 \longrightarrow 18\,H_2O + 16\,CO$

❖ However, the introduction of catalyst converter reduce the CO emissions

❖ Natural source of CO into the atmosphere is CH_4 gas



$2\,CH_4 + 3\,O_2 \longrightarrow 2\,CO + 4\,H_2O$

Methane is also produced in the stomach in the cattle and sheep and finally released into the atmosphere

36

Annadhasan L1-L2 · slide 36

Carbon Monoxide

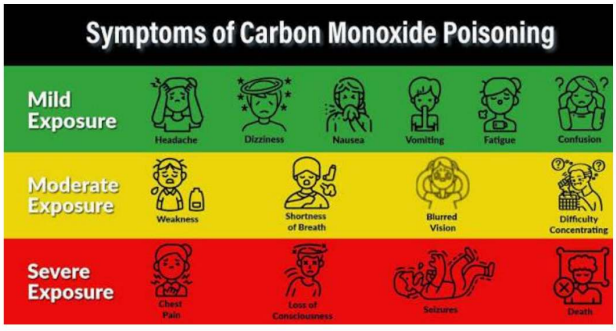
- Effect of CO on Human Health
  - Even the CO is most abundant air pollutant, it is not very toxic at the levels usually found in the atmosphere, however if allowed a build up in a confined space, it can cause serious health problem
  - CO interferes with the oxygen carrying capacity of blood; normally hemoglobin (Hb) in red blood cells combine with oxygen in the lungs to oxyhemoglobin (HbO_2). The HbO_2 is carried in the blood stream to various part of the body, where oxygen is released to the tissues
  - Carbon monoxide binds much more strongly to Hb than oxygen – Why?
  - If CO is present in the lungs, it displaces oxygen from Hb and thus reduces the amount of oxygen that can be delivered to the tissues

$HbO_2 + CO \longrightarrow HbCO + O_2$

- Treatment of CO poisoning:
- Symptoms of CO poisoning: headache, dizziness, impaired judgment, drowsiness, slowed reflexes, respiratory failure and death

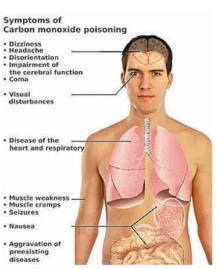
37

Annadhasan L1-L2 · slide 37



Annadhasan L1-L2 · slide 38

Carbon monoxide poisoning



35 ppm (0.0035%) Headache & Dizziness

200 ppm (0.02%) Headache & loss of judgement

3200 ppm (0.32%) Headache, Dizziness, Nausea

6400 ppm (0.64%) Respiratory arrest & death in less than 20 min

12800 ppm (1.28%) Unconsciousness. Death in less than 3 min

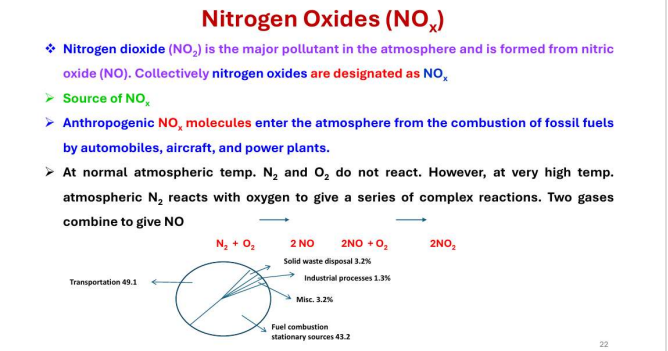
39

Annadhasan L1-L2 · slide 39

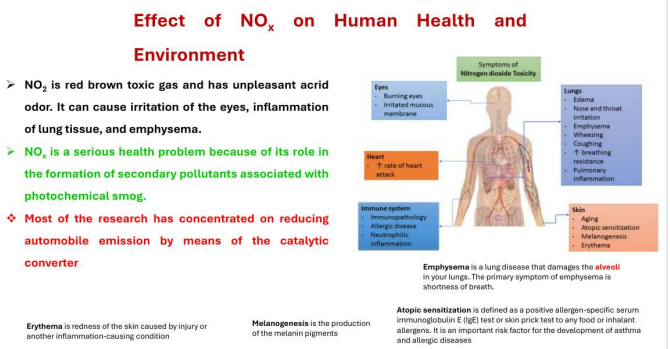
A5. Nitrogen oxides (L3-L5)

- NO_2 is the major pollutant and forms from NO; collectively NO_x. Sources: **transportation 49.1%**, stationary fuel combustion 43.2%, industrial 1.3%, solid waste 3.2%, misc 3.2%. Natural: bacterial decomposition of N-containing organic matter in soil.
- N_2 and O_2 do not react at normal temperatures; at very high temperature (engines, power plants): **$N_2 + O_2 \rightarrow 2NO$; $2NO + O_2 \rightarrow 2NO_2$**. NO has an unpaired electron (NO•).
- **NO_2 is red-brown, toxic, acrid**; it is the coloured oxide at room temperature ★ (N_2O_4 is colourless, forms on cooling). Effects: eye irritation, lung inflammation, emphysema; also erythema, melanogenesis, atopic sensitisation (definitions on slide). Role in secondary pollutants of photochemical smog.
- Fate: via OH, **$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$**; much HNO_3 forms in aqueous aerosols; droplets in clouds → acid rain. $HNO_3 + NH_3 \rightarrow NH_4NO_3$; nitrates fall as particles or in rain/snow (acid deposition).
- ⚠ The 2024 question 'which plant is badly affected by NO gas?' (options alfalfa/rice/lotus) is not answered in the slide text. Slides name alfalfa among SO_2-sensitive plants.

Slides from the course deck (Annadhasan L3-L5)



Annadhasan L3-L5 · slide 22

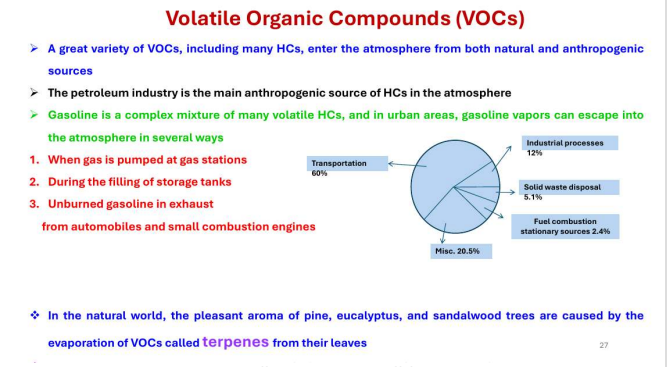


Annadhasan L3-L5 · slide 25

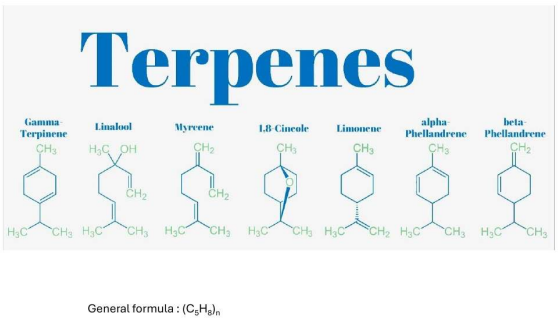
A6. VOCs, gasoline and engines (L3-L5)

- VOC sources: transportation 60%, industry 12%, solid waste 5.1%, stationary fuel 2.4%, misc 20.5%. Non-methane natural VOCs: **terpenes** $(C_5H_8)_n$: γ-terpinene, linalool, myrcene, 1,8-cineole, limonene, α- and β-phellandrene (pine, eucalyptus).
- **Gasoline** (% vol): alkanes 4-8, alkenes 2-5, isoalkanes 25-40, cycloalkanes 3-7, cycloalkenes 1-4, aromatics 20-50 (benzene 0.5-2.5). Tetraethyl lead was the anti-knock additive. Motor vehicles are major sources of CO, NO_x, HC.
- **Four-cycle engine**: limited O_2 → CO; very high T and P → NO; unburned HC. **Air/fuel ratio 14.7:1 is ideal ★**. Fuel-rich (lower A/F) → easier start, lower NO_x and HC but higher CO and worse mileage; air-rich → better efficiency but more NO_x.
- **Two-cycle engine** (scooters, mopeds, outboards): no valves, exhaust and fresh charge mix, so **25-30% of fuel leaves unburned**; fuel oil mix gasoline:oil ~40:1 (oil is C_{18}-C_{25}).
- Comparison (slide 33): 2-cycle outboard CO 165 g, NO_x 0.3 g, HC 89 g vs 4-cycle CO 127 g, NO_x 0.7 g, HC 7 g → 2-cycle emits >12× HC. Emission standards fell from 10.6 g/mi HC, 84 CO, 4.1 NO_x (pre-1970) to 0.41, 3.4, 0.4 (2000).

Slides from the course deck (Annadhasan L3-L5)



Annadhasan L3-L5 · slide 27



Annadhasan L3-L5 · slide 28

What is gasoline?

The typical composition of gasoline hydrocarbons (% volume) is as follows:
4-8% **alkanes**; 2-5% **alkenes**; 25-40% isoalkanes; 3-7% **cycloalkanes**; l-4% cycloalkenes; and 20-50% total **aromatics** (0.5-2.5% benzene)



Structure of Tetraethyl Lead

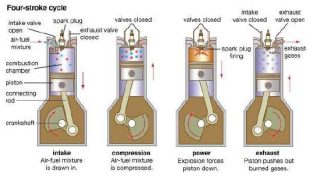
Tetraethyl lead acts as an anti knocking agent in gasoline and jet fuel and is added to petrol in order to reduce the ignition of vapors of petrol. Thus, tetraethyl lead is a petroleum additive.

Knocking occurs when fuel burns unevenly in your engine.

Annadhasan L3-L5 · slide 29

Volatile Organic Compounds (VOCs)

- **Automobile Four-Cycle Internal Combustion Engine**
  - Gasoline-powered four-cycle internal combustion engines are predominant in more developed countries
  - To know how NO_x, CO, and HCs are produced through this, we need to learn how this engine works
  - There are four steps in one complete cycle of the gasoline engine
  - In the confined space of the internal combustion engine, oxygen is in limited supply, and CO is formed and released to the atmosphere



30

Annadhasan L3-L5 · slide 30

Volatile Organic Compounds (VOCs)

- **Automobile Four Cycle Internal Combustion Engine**
  - In the internal combustion engine, temp and pressure are very high, and NO is formed from N_2 and O_2. The unburned HCs are emitted with the burned gas as pollutant VOCs
  - The air/fuel ratio has a dramatic effect on the emission of pollutants from the four-cycle engine
  - During the engine tune-up, adjustment of the carburetor or the fuel injection system can greatly reduce the emission of pollutants.
  - The ideal stoichiometric air/fuel ratio is 14.7:1
  - Adjustment to a more fuel-rich (lower air/fuel) ratio makes the car easier to start and decreases the emission of NOx and HCs; unfortunately, it also decreases the fuel efficiency, and the car will get fewer Km per liter. The emission of CO is also increased as air is less available
  - Adjustment to a more air-rich (higher air/fuel) ratio will make the car harder to start; although it increases fuel efficiency, the emission of NO_x is increased
  - The ideal ratio is a compromise between fuel efficiency and the emission of the lowest collective amount of pollutant

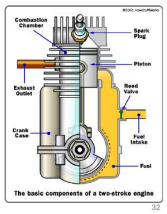
31

Annadhasan L3-L5 · slide 31

Volatile Organic Compounds (VOCs)

- **Gasoline Powered Two-Cycle Engine**
- A two-cycle engine is used to power motor scooters, mopeds, and snowmobiles. It is less complicated than the four-cycle engine and also less expensive.
- It takes fuel, releases exhaust in the same stroke, and emits from 25% to 30% of the fuel consumed as unburned HCs.
- The two-cycle engine has no valves, and that power is produced in one of two steps rather than one in four steps; this means the two-cycle engine runs faster and hotter than the four-cycle engine

1. In two-cycle engine motor oil: gasoline 1: 40 ratio
2. Motor oil has C_{18} to C_{25} HCs i.e., higher molecular weight
3. In the two-cycle engine, combustion takes place at a lower temperature than the four-cycle engine
4. Due to higher molecular weight, motor oil is not vaporized and burned efficiently like gasoline
5. Two step cycle requires that the exhaust gases from combustion leave the cylinder through the exhaust port while the fresh air/fuel mixture for the next power stroke is simultaneously enters the cylinder from the side arm



32

Annadhasan L3-L5 · slide 32

Volatile Organic Compounds (VOCs)

- **Comparison**

1. A comparison of the emission from marine outboard two- and four-cycle engines of the same horsepower was made. The result of those tests show that the exhaust of a two-cycle engine contained more than 12 times the amount of HCs than a four-cycle engine of the same power

Type of Marine outboard Engine	CO(g)	NO_x(g)	HC(g)
Two cycle Engine	165	0.3	89
Four Cycle Engine	127	0.7	7

Source: Juttner F. D. et al., Water Research **1995**, *29*, 1976-1982


33

Annadhasan L3-L5 · slide 33

Automobile Pollutants

- Motor vehicles are a major source of CO, NO_x, and volatile HCs
- Since 1975, when all new cars in the USA were required by law to be equipped with a catalytic converter, emissions of those pollutants have been reduced significantly
- The table shows that today's car emits 95% less pollutants than pre-1970 vehicles despite the fact that the number of miles traveled has almost doubled in the last 20 years

Emission Standards for Light duty Vehicles

	Federal Standards				California Standards			
Year	HCs	CO	NOx	Evaporative HCs	HCs	CO	NOx	Evaporative HCs
<1970	10.6	84	4.1	> 45	10.6	84	4.1	>45
1970	4.1	34	-	-	4.1	34	-	2
1975	1.5	15	3.1	2	0.9	9	2.0	2
1980	0.41	7.0	2.0	2	0.39	9	1.0	2
1985	0.41	3.4	1.0	2	0.39	7	0.4	2
1990	0.41	3.4	1.0	2	0.39	7	0.4	2
1993	0.41	3.4	1.0	2	0.25	3.4	0.4	2
2000	0.41	3.4	0.4	2	0.25	3.4	0.4	2

All values reported in gms per mile except for evaporative HCs which are expressed as gms per tests

34

Annadhasan L3-L5 · slide 34

A7. Catalytic converter (L3-L5)

- **Three-way catalytic converter**: ceramic honeycomb coated with **Pt, Pd, Rh**; removes CO, HC and NO_x together. Two chambers in succession.
- **Chamber 1 (reduction), Rh ★**: HC + H_2O → H_2 + CO; 2NO + 2H_2 → N_2 + 2H_2O. Rh reduces NO_x to N_2; it does not oxidise CO.
- **Chamber 2 (oxidation), Pt and Pd**, with air injected: 2CO + O_2 → 2CO_2; HC + 2O_2 → CO_2 + 2H_2O.
- **Overall: 2NO + 2CO → N_2 + 2CO_2** (Rh, Pt, Pd). Other reactions listed: C + 2NO_2 → CO_2 + 2NO; CO + NO → CO_2 + ½N_2; 2CO + NO_2 → 2CO_2 + ½N_2; H_2 + NO → H_2O + ½N_2; C + O_2 → CO_2; general HC combustion.
- Needs A/F near 14.7:1 and warm operating temperature (~0 conversion cold). Market vendors: BASF SE, CDTi, Continental AG, Eberspächer, Faurecia. Slide tag: 'Thermoelectric materials'.
- ⚠ The 2024 question 'Rh in the two-way catalytic converter converts NO to ...?' → Rh reduces NO to N_2 (answer is the NO → N option, not HC → CO).

Slides from the course deck (Annadhasan L3-L5)

Catalytic Converter

Reduction of nitrogen oxides to nitrogen (N_2)

- $C + 2NO_2 \rightarrow CO_2 + 2NO$
- $CO + NO \rightarrow CO_2 + \frac{1}{2}N_2$
- $2CO + NO_2 \rightarrow 2CO_2 + \frac{1}{2}N_2$
- $H_2 + NO \rightarrow H_2O + \frac{1}{2}N_2$

Oxidation of carbon, hydrocarbons, and carbon monoxide to carbon dioxide

- $C + O_2 \rightarrow CO_2$
- $CO + \frac{1}{2}O_2 \rightarrow CO_2$
- $a\,C_xH_y + b\,O_2 \rightarrow c\,CO_2 + d\,H_2O \qquad a, b, c, d, x, y \in \mathbb{Z}$

Thermoelectric materials



Catalytic Converter Operation

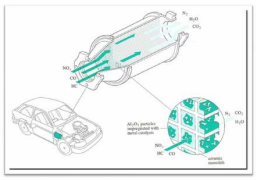


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The Catalytic Converter

- The use of a three-way catalytic converter (used since 1981) can reduce the emission drastically
- It is called a three-way catalytic converter as it simultaneously reduces the amounts of HCs, NO_x, and CO in the exhaust stream
- The converter is a very fine honeycomb structure made of ceramic coated with precious metals like Pt, Pd, and Rh, which act as catalysts





35

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Top Five Automotive Catalytic Converter Market Vendors

- BASF SE — BASF, We create chemistry
- CDTi Advanced Materials Inc. — CDTi Advanced Materials, Inc.
- Continental AG — Continental
- Eberspacher Group — Eberspächer
- Faurecia SA — faurecia, inspiring mobility

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The Catalytic Converter

- The catalytic converter has two chambers in succession. As the gases enter, **Rh** catalyzes the reduction of NO_x to nitrogen gas by hydrogen generated at the surface of the **Rh** catalyst by the reaction of H_2O and unburned HCs

$$HCs + H_2O \longrightarrow H_2 + CO\,; \qquad 2NO + 2H_2 \longrightarrow N_2 + 2H_2O$$

- Then air is injected into the exhaust stream to introduce oxygen and **CO** is oxidized to **CO_2** in the presence of **Pt** and **Pd** catalyst

$$2\,CO + O_2 \longrightarrow 2\,CO_2\,; \quad HC + 2O_2 \longrightarrow CO_2 + 2\,H_2O$$

- Thus, overall reaction for reduction of NO and oxidation of CO can be written as

$$2\,NO + 2\,CO \xrightarrow{Rh,\,Pt,\,Pd\ Cat} N_2 + 2\,CO_2$$

37

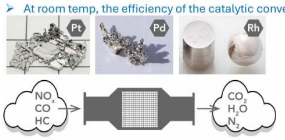
Annadhasan L3-L5 · slide 37

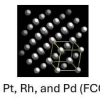
The Catalytic Converter

- Oxidation of a typical gasoline HC, Octane occurs as

$$2\,C_8H_{18} + 25\,O_2 \xrightarrow{Pt,\,Pd\ Cat} 16\,CO_2 + 18\,H_2O$$

- Automobiles using catalytic converters must have their air/fuel ratio set as 14.7:1 to get optimum results.
- At room temp, the efficiency of the catalytic converter is nearly zero





Surface chemistry

Pt, Rh, and Pd (FCC)

38

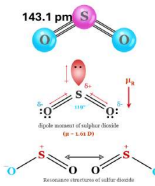
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A8. Sulfur dioxide and acid rain (L3-L5)

- SO_2 (S-O 143.1 pm) is the main cause of acid rain. Fossil fuels ~70%, industry ~23%; coal contains pyrite (FeS_2). Reactions: $S + O_2 \rightarrow SO_2$; $2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O$ (H_2S from anaerobic decay).
- Natural: volcanic eruptions (**Pinatubo, June 1991, 25 Mt SO_2**) forming H_2SO_4 aerosols that cool the planet.
- **Residence time 10-30 days**. Fate: $SO_2 + ½O_2 \rightarrow SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$; aqueous: $SO_2 + H_2O \rightleftharpoons H_2SO_3$; $HSO_3^- + H_2O_2 \rightarrow HSO_4^- + H_2O$; $HSO_3^- + O_3 \rightarrow HSO_4^- + O_2$. Cloud-water pH can fall to ~3 (acid rain).
- **Oxoacids of sulfur ★** (slide 43): sulfurous H_2SO_3; sulfuric H_2SO_4; thiosulfuric $H_2S_2O_3$; dithionous $H_2S_2O_4$; **dithionic $H_2S_2O_6$**.
- Effects: SO_2 is colourless, pungent; irritates eyes/respiratory tract (children, asthmatics); harms barley, alfalfa, cotton, wheat; corrodes marble and limestone (Taj Mahal yellowing, slide 44).
- Control: remove sulfur before burning (coal washing) or after: **flue gas desulfurisation** with limestone/dolomite slurry, **$2SO_2 + 2CaCO_3 + O_2 \rightarrow 2CaSO_4 + 2CO_2$** (gypsum).
- CS_2 (S=C=S, 155.26 pm): colourless liquid, ether-like odour, bp 46.3 °C; $CS_2 + 3O_2 \rightarrow CO_2 + 2SO_2$ (slide writes unbalanced form).

Slides from the course deck (Annadhasan L3-L5)

Sulfur Dioxide



- The release of SO_2 into the atmosphere is the primary cause of acid rain
- Fossil fuel combustion accounts for 70% of the emission
- Industrial sources contribute approximately 23% of SO_2
- Coal oil and all other fossil fuels naturally contain some sulfur (FeS).

- When sulfur containing coal is burned, the sulfur is oxidized to SO_2: $S + O_2 \longrightarrow 2SO_2 + 2H_2O$
- H_2S (produced as an end product of anaerobic decomposition of sulfur containing organic matter by microorganism) enter to the atmosphere to form SO_2

$$2H_2S + 3O_2 \longrightarrow 2SO_2 + 2H_2O$$

39

Annadhasan L3-L5 · slide 39

Sulfur Dioxide





- Volcanic eruptions are another more localized natural source of SO_2; e.g., the eruption of Mt. Pinatubo in the Philippines in June 1991 contributed 25 million tons of SO_2 into the atmosphere, where it was converted into sulfuric acid aerosols

Fate of atmospheric SO_2: Acid Rain

- The residence time of SO_2, controlled not only by chemical conversion to sulfate but also by dry deposition, is 10-30 days.
- SO_2 in the atmosphere reacts with oxygen to form SO_3, which then readily reacts with water vapor or water

Annadhasan L3-L5 · slide 40

The fate of atmospheric SO_2: Acid Rain

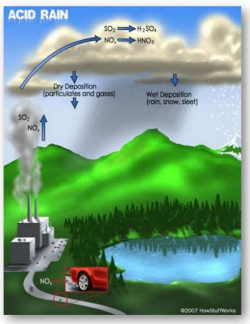
$$SO_2 + OH\cdot \longrightarrow HSO_3\cdot$$
$$HSO_3\cdot + O_2 \longrightarrow SO_3 + HOO\cdot$$
$$SO_3 + H_2O \longrightarrow H_2SO_4\ (g)$$
$$H_2SO_4 + H_2O \longrightarrow H_2SO_4\ (aq)$$

- Sulfuric acid in the atmosphere becomes concentrated near the base of clouds where the pH level is as low as 3 (pH of orange juice)
- Some of the atmospheric SO_2 dissolves if there is significant water in the air to form sulfurous acid

$$SO_2(g) + H_2O\ (aq) \longrightarrow H_2SO_3(aq)$$

- The dissolved SO_2 is oxidized by trace amount of H_2O_2 and Ozone that are also present in the aerosol droplets to sulfate ions (SO_4^{2-})
- $HSO_3^- + H_2O_2 \longrightarrow H_2O + HSO_4^-$

$$HSO_3^- + O_3 \longrightarrow O_2 + HSO_4^-$$



41

Annadhasan L3-L5 · slide 41

The fate of atmospheric SO_2: Acid Rain

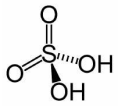
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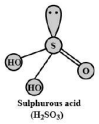
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$$HSO_3^- + O_3 \longrightarrow O_2 + HSO_4^-$$



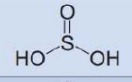
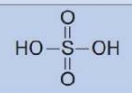
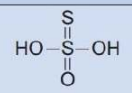
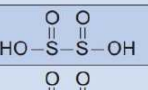
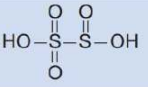
H_2SO_4 *vs* H_2SO_3



42

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Oxoacids of Sulfur

Name	Molecular Formula	Structure
Sulphurous acid	H_2SO_3	
Sulphuric acid	H_2SO_4	
Thiosulphuric acid	$H_2S_2O_3$	
Dithionous acid	$H_2S_2O_4$	
Dithionic acid	$H_2S_2O_5$	

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Effect of Acid Rain



Original color of Taj Mahal

Taj Mahal after the constant effect of acid rain

44

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Effect of SO_2 on Human Health and Environment

- SO_2 is a colorless, toxic gas with a sharp, acidic odour
- Exposure to it causes irritation of eyes, respiratory passages, and aggravates symptoms of respiratory disease. Children are susceptible to its effects
- SO_2 is also harmful to plants. Crops such as barley, alfalfa, cotton, and wheat are particularly adversely affected


Barley

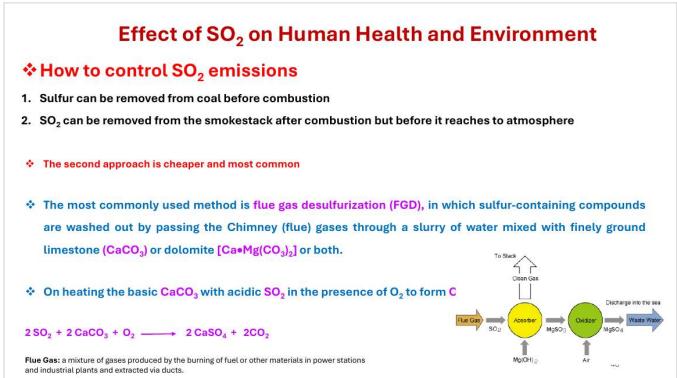

Alfalfa


Cotton

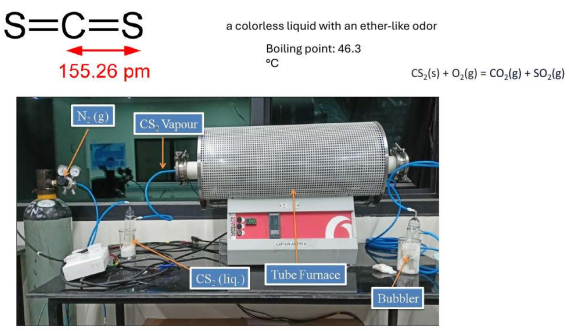

Wheat

45

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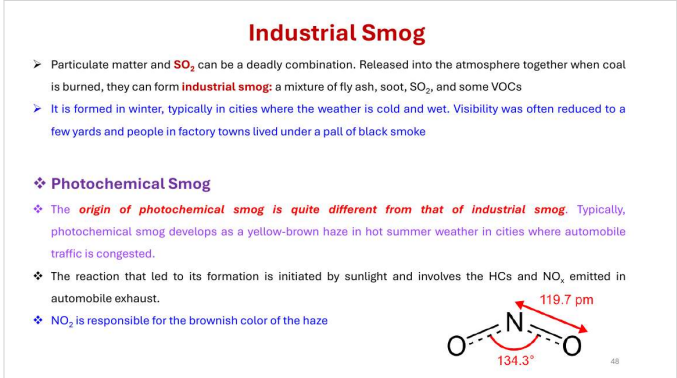
Annadhasan L3-L5 · slide 47

A9. Smog (L3-L5)

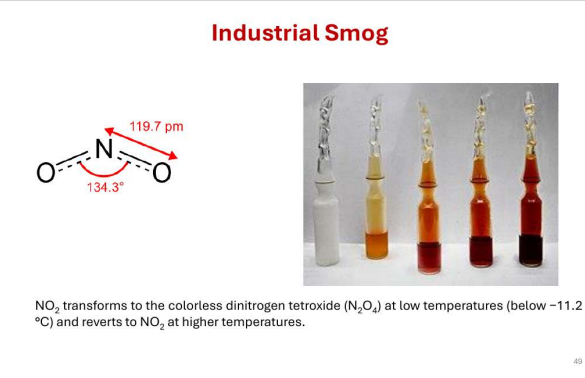
	Industrial (London-type) smog	**Photochemical (Los Angeles-type) smog**
Weather	Cold, wet, winter; fog	Hot, dry, sunny; summer
Ingredients	SO_2 + particulates (soot, fly ash) from coal; NO_2 can form colourless N_2O_4 in cold air	NO_x + hydrocarbons + sunlight
Look	Grey/black fog	Brown/yellow haze (NO_2)
Key species	SO_2, H_2SO_4 aerosol, PM	O_3, PAN, aldehydes, HNO_3

- Particulate matter and SO_2 together can be deadly (London 1952).
- **Hydroxyl radical production**: $NO_2 + h\nu$ (<320 nm) → NO + O; $O + O_2 → O_3$; $O_3 + NO → NO_2 + O_2$. $O_3 + h\nu → O^* + O_2$; $O^* + H_2O$ → **2OH•** (one NO_2 → two OH). Products of radical-radical reactions are very soluble in water and are rained out.
- HC oxidation: $RH + OH• → R• + H_2O$; $R• + O_2 → RO_2•$; $RO_2• + NO → RO• + NO_2$; leading to aldehydes and $HO_2•$; $HO_2• + NO → OH• + NO_2$. **Each hydrocarbon molecule can generate four OH radicals ★** (2024 options 10,000/100/10/...; chain reaction).
- **Reactivity (C-H bond energy, kJ/mol)**: CH_3-H 427 (methane) · C_2H_5-H 406 · C_3H_7-H 393 · $HOCH_2$-H 393 · C_6H_5-H 427 (benzene) · **toluene benzylic C-H 326 (weakest, fastest)**. Weaker bond → faster H-abstraction; 3° > 2° > 1°.
- **OH addition to C=C** (alkenes) is faster than abstraction; gives hydroxy-radical adducts → acetaldehyde, formaldehyde etc.
- **Secondary smog reactions**: aldehydes + OH → acyl radical → peroxyacyl radical + NO_2 → **PAN** ($RCOO_2NO_2$, e.g. $CH_3COO_2NO_2$), the major **eye irritant ★**; O_3 is the most abundant harmful oxidant (eyes, nose, plants).
- Indoor pollution: air inside can be worse because of poor circulation (slide 57).

Slides from the course deck (Annadhasan L3-L5)

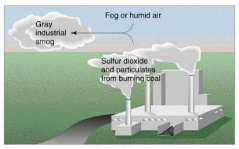


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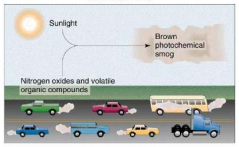


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Industrial Smog



(a) Industrial smog



(b) Photochemical smog





50

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Production of Hydroxyl Radicals

- By the time sunlight reaches the surface of the Earth, all of the light energy UV light has been absorbed in the stratosphere
- NO_2 is the only automobile emission that is capable of absorbing visible light that reaches the Earth's surface
- NO_2 + sunlight (< 320 nm) $\longrightarrow$ NO + O ; O + O_2 $\longrightarrow$ O_3
- O_3 + NO $\longrightarrow$ NO_2 + O_2

- The ozone produced absorbs light in the blue region of the visible spectrum (< 320 nm) and photo dissociates: $O_3 \longrightarrow O_2 + O$
- The oxygen atom produced (having six electrons) reacts with water vapor in the atmosphere and abstracts a hydrogen atom to produce hydroxyl radical (seven electrons)
- $O + H_2O \longrightarrow 2OH^\bullet$ By this way one NO_2 molecule produces two hydroxyl radicals
- The concentration of hydroxyl radicals does not continue to increase out of control because there are termination reactions that remove it from the troposphere. It can react with other radical species in the troposphere

51

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Production of Hydroxyl Radicals

$^\bullet OH + {}^\bullet NO_2 \longrightarrow HNO_3$

$^\bullet OH + HOO^\bullet \longrightarrow H_2O + O_2$

$2\,{}^\bullet OOH \longrightarrow H_2O_2 + O_2$

❖ The products are very soluble in water and are removed from troposphere during precipitation

- Unburned HCs in automobile exhausts (RCH_3) react with hydroxyl radicals to form a number of secondary pollutants, including HC radicals $RO_2^\bullet$. This radical then reacts with NO to form aldehydes and the hydroperoxide radical $HO_2^\bullet$

$RCH_3 + {}^\bullet OH \longrightarrow {}^\bullet RCH_2 + H_2O$

$RCH_2^\bullet + O_2 \longrightarrow RCH_2OO^\bullet$ (Peroxy alkyl radical)

$RCH_2OO^\bullet + NO \longrightarrow RCH_2O^\bullet$ (Alkoxy radical)

$RCH_2O^\bullet + O_2 \longrightarrow RCHO + {}^\bullet OOH$ (Hydroperoxyl radical)

$HOO^\bullet + NO \longrightarrow NO_2 + {}^\bullet OH$

52

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Production of Hydroxyl Radicals

❖ Each step in this reaction produces a radical. The overall reaction can be summarized as

$RCH_3 + 2\,O_2 + 2\,NO \longrightarrow RCHO + 2\,NO_2 + H_2O$ (1)

$2\,NO_2 + 2\,H_2O \longrightarrow 2\,NO + 4\,{}^\bullet OH$ (2)

$RCH_3 + 2\,O_2 + H_2O \longrightarrow RCHO + 4{}^\bullet OH$

❖ This reaction produces four hydroxyl radicals for every HC reacted. This is a catalytic reaction. A very small number of radicals can produce a large amount of product through the production of four radicals per cycle

Reaction of Hydroxyl Radicals with HCs

❖ Abstraction of Hydrogen

- Hydroxyl radicals will react with certain unburned HCs from the automobile exhaust depending on the number and type of C-H bonds in the HCs

53

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Reaction of Hydroxyl Radicals with HCs

Compound	Bond	Energy KJ/mol
Methane	CH_3-H	427
Ethane	CH_3-CH_2-H	406
Propane	CH_3-CH_2-CH_2-H	393
Methanol	$HOCH_2$-H	393
Benzene	H_5C_6-H	427
Toluene	$H_5C_6H_2C$-H	326

- The dissociation reaction is R-H $\longrightarrow$ R• + H•
- The bond dissociation energy depends on how the R radical is stabilized $3^o > 2^o > 1^o$ i.e., a tertiary C-H bond more easily dissociate than a secondary C-H bond
- For benzene, the C-H has high dissociation energy as the aromatic carbon is sp^2, which mean sp^2 C-H bonds are much stronger than aliphatic C-H bond due to higher *s*-character
- Higher reaction rates (low dissociation energy) are observed for toluene or xylene as the $C_6H_5CH_2$• can be stabilized by adjacent aromatic ring

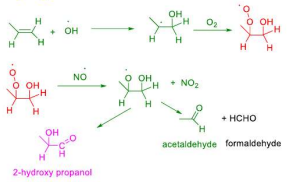
54

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Reaction of Hydroxyl Radicals with HCs

❖ **Addition to the double bond**

- **The reaction of hydroxyl radicals with alkenes proceeds at even a faster rate than the hydrogen abstraction for HCs**
- **This reaction is not a hydrogen atom extraction but rather an addition of the hydroxyl radical to the double bond**



- **The NO_2 produced in this reaction can go on to make more ozone, and the products of this addition reaction i.e., acetaldehyde, formaldehyde, and 2-hydroxypropanol all go on to form other pollutants in secondary smog-forming reactions**

55

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Secondary Smog Forming Reactions

- The products formed in the previous reactions can undergo further reactions to the troposphere. The following reaction sequence takes place for all aldehydes formed. Acetaldehyde is used as example

$CH_3CHO + {}^\bullet OH \longrightarrow CH_3CO^\bullet + H_2O$

$CH_3CO^\bullet + O_2 \longrightarrow CH_3COOO^\bullet$

$CH_3COOO^\bullet + {}^\bullet NO_2 \longrightarrow CH_3COOONO_2$

PAN

$H_3C-\overset{O}{\overset{\|}{C}}-O-O-NO_2$

Peroxyacetyl nitrate

- PAN is the component of smog that causes major eye irritation. PANs are relatively stable molecules and have long lifetimes in cooler air
- Ozone, aldehyde, and PANs all contribute to the harmful effects of photochemical smog, but ozone the pollutant produced in the greatest quantity causes the most serious problems in the troposphere
- Ozone in the stratosphere protects us from damaging UV radiation from the Sun
- Ozone is a powerful oxidizing agent and causes irritation in the eyes and nasal passages
- Ozone is also very toxic to plants

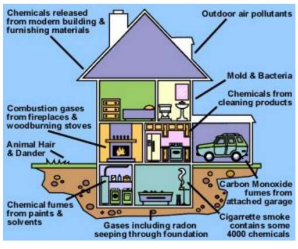
56

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Indoor Pollution

- One can expect to be safer from air pollutants indoors, but in today's well-sealed homes and offices, this is often not the case
- In buildings where there is little or no circulation or fresh air, pollutants may accumulate to a dangerous level

Lead pipes



57

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A10. Stratospheric ozone (L3-L5)

- Ozone layer ~**20-30 km** altitude; absorbs UV; 90% of ozone is in the stratosphere. **1 Dobson unit = 2.7×10^{16} O_3 molecules per cm^2** ★ (a 0.01 mm layer at STP). Average ~293.4 DU in 2000; hole when < ~220 DU.
- **Chapman cycle**: O_2 + hν (**<240 nm**) → 2O, ΔH = 498 kJ/mol; O + O_2 + M → O_3 + M (heat); O_3 + hν (220-330 nm) → O_2 + O; O + O_3 → 2O_2. Dynamic equilibrium; the sum of the first two steps is heat released.
- UV classes: UV-C 250-290 nm (absorbed by stratospheric O_3), UV-B 290-320 nm (O_3; skin damage), UV-A 320-380 nm (reaches surface; NO_2 absorbs), far UV 10-240 (O_2, N_2), visible 400-750.
- **Catalytic destruction**: X + O_3 → XO + O_2; XO + O → X + O_2; **net O_3 + O → 2O_2**. X = OH, NO, Cl, Br.
- **OH cycle**: OH from O + H_2O, O + CH_4, H_2O + hν; ~half of lower-stratosphere loss; **~40 cycles per OH**. **NO cycle**: N_2O (inert, from soil bacteria, fertilisers, combustion) reaches the stratosphere: N_2O + O* → 2NO; **~10,000 O_3 per NO_x**. **Cl cycle**: CFCs (refrigerants, propellants, solvents; non-toxic, non-flammable, lifetime >100 y) photolyse to Cl•; **~100,000 O_3 per Cl**; **Br is more efficient than Cl** (halons, fire extinguishers); Rowland and Molina, 1974.

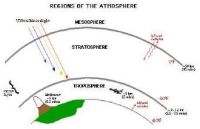
Slides from the course deck (Annadhasan L3-L5)

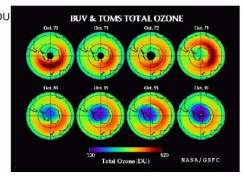
Chemistry of the Stratosphere

- Today, the ozone layer is considered the Earth's natural sun-screen because it filters harmful UV radiation before it can reach the surface of the Earth
- A substantial reduction in the amount of ozone in the ozone layer could threaten all life on Earth

Dobson Unit

- The Dobson Unit (DU) is used to describe the amount of ozone in the stratosphere
- It is named after Gordon M. B. Dobson, who built the first instrument to measure the total abundance of ozone from the ground
- **1 DU = 2.7 x 10^{16} ozone molecules/sq centimeter**
- In the past, the average amount of ozone covering the Earth was more than 270 DU
- In 2000, the average abundance of ozone from 90° N to 90°S was 293.4 DU
- However, the amount of ozone over the Antarctic is 220 DU which is refer to the ozone hole





58

Annadhasan L3-L5 · slide 58

The production of Ozone in the Stratosphere

- In the absence of pollution, ozone (O_3), is not present in any appreciable amount in the troposphere but occurs naturally in the stratosphere
- Ozone's concentration is greatest at an altitude of 20 to 30 km from the Earth's surface
- The ozone layer is formed when an ordinary molecule of oxygen gas (O_2) in the stratosphere absorbs UV radiation from the sun wavelength less than 240 nm and this dissociates O_2 to a single oxygen atom (O)

O_2 + hv (< 240 nm) ⟶ O + O (slow) ΔH = 498 KJ/mol



- Single oxygen atoms are very reactive and immediately combine with O_2 to form O_3:
- O + O_2 + M ⟶ O_3 + M + Heat (fast)
- M can be a third molecule (N_2) present in the atmosphere
- The heat generated by this reaction is carried away by the third molecule
- At the same time, the ozone formed absorbs UV radiation very strongly with wavelengths of 220-330 nm to be broken down into an oxygen molecule and oxygen atom
- Also, when O_3 encounters an oxygen atom, two can combine to form two O_2 molecules
- A dynamic equilibrium is established

59

Annadhasan L3-L5 · slide 59

The production of Ozone in the Stratosphere

Common names for the UV spectrum			
l(nm)	Name	Species absorbing	Location
10-240	Far UV	O_2, N_2	Thermosphere
250-290	UV-C	O_3	Stratosphere
290-320	UV-B	O_3	Stratosphere, troposphere
320-380	UV-A	NO_2	Polluted troposphere
400-750	Visible	Many	Earth's Surface

Catalytic destruction of Ozone

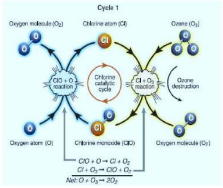
- Ozone is a very reactive molecule and is termed a meta-stable molecule i.e. it decomposes slowly in contact with a molecule of another gas
- There are a number of species that react efficiently by abstracting an oxygen atom from the ozone molecule. Say "X" designates the reactive species. Thus, the reaction steps are
  - i) X + O_3 ⟶ XO + O_2; ii) XO + O ⟶ X + O_2
- These are the additional steps for ozone destruction compared to the steps discussed earlier

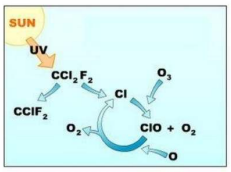
60

Annadhasan L3-L5 · slide 60

The production of Ozone the Stratosphere

- So, the sum of above two reactions is O_3 + O ⟶ 2O_2
- The overall reaction does not contain the X species, because X is not consumed in the reaction. It acts as a catalyst for the destruction of ozone
- "X" lowers the activation energy and atomic oxygen is required to regenerate X
- The catalytic species "X" can be i) Hydroxyl radical, ii) nitric oxide, iii) Chlorine , and iv) bromine atoms





DESTRUCTION OF OZONE BY CFC

61

Annadhasan L3-L5 · slide 61

Catalytic destruction of Ozone

Hydroxyl Radical Cycle

- Hydroxyl radical (•OH) can be produced by two different photochemical processes. The first is hydrogen abstraction with either water or methane
  - i) O + H_2O ⟶ 2 •OH ; ii) O + CH_4 ⟶ •OH + •CH_3
- The second is photolysis of **water** H_2O + hv ⟶ H• + •OH
- Hydroxyl radical is responsible for nearly one-half of the total ozone destruction in the lower stratosphere

$^{\bullet}OH + O_3 \longrightarrow {}^{\bullet}OOH + O_2$

$^{\bullet}OOH + O \longrightarrow {}^{\bullet}OH + O_2$

One •OH radical can have as many as 40 cycles i.e., 40 ozone molecule can decompose

Net reaction $O + O_3 \longrightarrow 2O_2$

62

Annadhasan L3-L5 · slide 62

Catalytic destruction of Ozone

Nitric Oxide Cycle

- Auto mobiles and truck engines release large quantities of nitric oxide (NO) into the troposphere
- Nitric oxide is converted to NO_2 and eventually to HNO_3 which fall down as rain fall before reaching to stratosphere
- N_2O is much less reactive than NO and reaches to stratosphere
- Above 30 km, the N_2O can absorb high energy photons to produce molecular nitrogen and an excited oxygen atom (O*) N_2O + hv ⟶ N_2 + O*
- Below 30 km in the stratosphere, the excited state oxygen reacts with the N_2O to produce NO
  - ii) N_2O + O* ⟶ 2 NO*
- NO can act as X in the catalytic process

$NO + O_3 \longrightarrow NO_2 + O_2$

$NO_2 + O \longrightarrow NO + O_2$

Net $O + O_3 \longrightarrow 2O_2$

1 NOx radical can catalyze the destruction of 10,000 ozone molecules

63

Annadhasan L3-L5 · slide 63

Catalytic destruction of Ozone

The Chlorine Cycle

- The major sources of stratospheric chlorine and bromine have been the anthropogenic **chlorofluoro carbons (CFCs)** and bromine-containing halons (haloalkanes)
- CFCs used as refrigerants, propellants for aerosol spray, and solvents for cleaning electric circuit
- Halons are used in commercial fire extinguisher system
- CFCs and halons are nontoxic and nonflammable, which make them superior to other gases that were used for the same application
- CFCs are so unreactive they don't break down when released and can persist in the troposphere for more **than 100 years.** Over time, air currents carry them into the stratosphere
- In 1974, two chemists, F. S. Rowland and Mario Molina, predicted that when exposed to UV radiation in stratosphere, CFCs would break down to form chlorine radical ($Cl^{\bullet}$)
- Each chlorine radical involved in a chain reaction has the potential to destroy 100,000 molecules of ozone before winds carry it back to the troposphere
- Bromine-containing compounds are more efficient in destroying the ozone

64

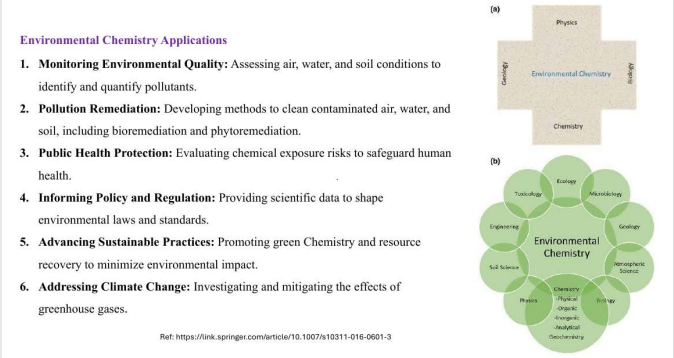
Annadhasan L3-L5 · slide 64

Part B: Dr. Kishore Natte — Organic chemicals in the environment

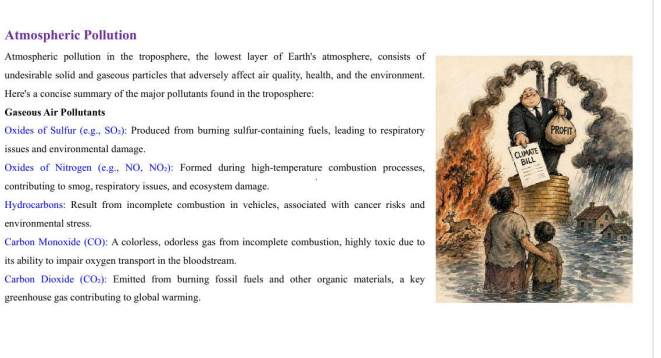
B1. Framing, economy models and waste (Class 1-3)

- Environmental chemistry studies sources, reactions, transport, effects and fate of chemicals. Applications: monitoring, remediation (bio- and phytoremediation), public health, policy, sustainable practice, climate.
- **Green chemistry** (EPA): design of products and processes that reduce or eliminate hazardous substances across the life cycle. **Green engineering**: reduce pollution and promote sustainability while remaining economically viable. **Sustainable development**: meet present needs without compromising future generations.
- Atmospheric pollution: troposphere (smog, particulates), stratospheric ozone loss, greenhouse gases. Organic pollutants: **VOCs** (benzene, thinners) and **POPs** (DDT, industrial chemicals). Slides on plastic pollution, pesticides, ozone depletion.
- Plastics last long: bags, bottles 20-500 y (Class 1-3) ⚠; single-use plastic examples.
- **Linear economy**: take-make-dispose. **Recycling**: used material → new products. **Downcycling**: lower-quality product (artificial turf filler example). **Circular economy**: regenerative, closed loop, reuse and repair, design for longevity, designing out waste.
- Pharmaceutical waste: expired/unused drugs, HPAPIs, controlled drugs, empty containers; principles: reduce, reuse, reprocess, renewable, recyclable.

Slides from the course deck (Kishore Class 1-3)



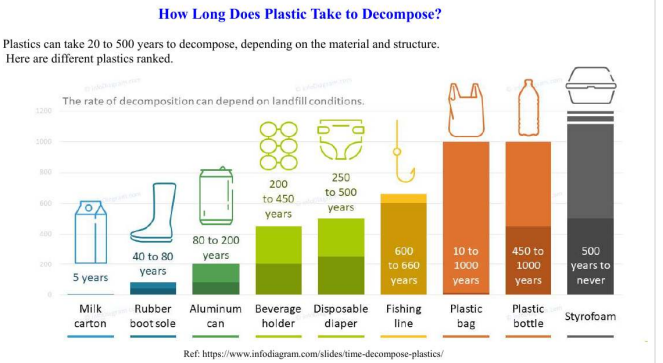
Kishore Class 1-3 · slide 3



Kishore Class 1-3 · slide 4



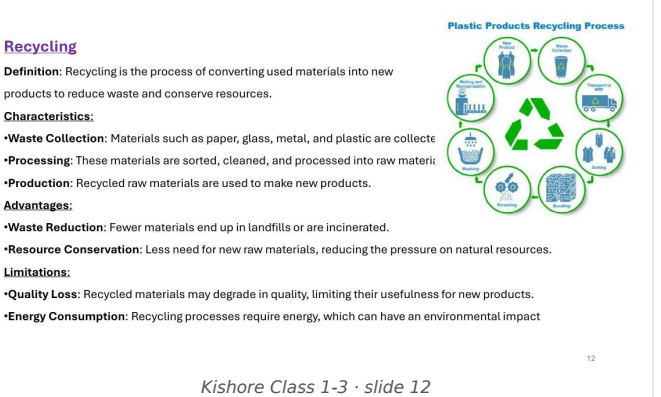
Kishore Class 1-3 · slide 6



Kishore Class 1-3 · slide 7



Kishore Class 1-3 · slide 11



Kishore Class 1-3 · slide 12

Downcycling



Definition: Downcycling is a form of recycling where the recycled material is converted into products of lower quality and functionality than the original material.

Characteristics:

- **Quality Reduction**: Materials lose quality with each recycling cycle.
- **Limited Applications**: Recycled materials are often used for products with lower performance requirements.
- **Example**: Artificial grass can often only be converted into lower-quality products like filler material or secondary construction materials after use.

Advantages:

- **Waste Reduction**: Although quality decreases, the material stays in use longer and out of the waste stream.
- **Resource Conservation**: Better than disposal, but not as effective as high-quality reuse.

Limitations:

- **Quality Loss**: The material is no longer suitable for high-quality applications, reducing its circular value.
- **Temporary Solution**: Eventually, the material often ends up as waste.

13

Kishore Class 1-3 · slide 13

Circular Economy



Definition: A circular economy is a regenerative system where resources, materials, and products are kept in use for as long as possible, reused, and recycled to minimize waste and environmental impact.

Characteristics:

- **Design for Sustainability**: Products are designed for longevity, reuse, and easy recycling.
- **Closed Loop**: Materials circulate within the system without losing value.
- **Reuse and Repair**: Products and components are reused and repaired instead of discarded.

Advantages:

- **Minimal Waste Production**: By reusing and recycling materials, the amount of waste is greatly reduced.
- **Value Retention**: Materials retain their value and utility within the system, which is economically beneficial.
- **Sustainable Resource Use**: Reduces the need for new raw materials and minimizes environmental impact.

Challenges:

- **Infrastructure and Systems**: Requires new infrastructures and economic models.
- **Behavioral Change**: Requires a change in how businesses and consumers handle products and materials.

14

Kishore Class 1-3 · slide 14

B2. POPs, pesticides and synthetic organic chemicals (Class 1-3)

- Organic compounds: carbon-based; **'organic' does not mean natural or safe** ★ (risk depends on toxicity, concentration, exposure route, duration). ~80,000 synthetic chemicals are in use, most untested. SOCs: man-made carbon-based compounds in pesticides, solvents, plastics, pharmaceuticals.
- **POPs**: resist chemical, biological and photolytic breakdown; travel by wind and water (Arctic contamination, 'grasshopper effect'); fat-soluble, water-insoluble → bioaccumulate and biomagnify.
- Terms: **insecticide** (insects), **pesticide** (any pest incl. weeds, fungi, rodents), **herbicide** (plants), **fungicide** (fungi).
- Why pesticides: disease vectors, e.g. plague (*Yersinia pestis*, fleas/rats, Black Death 1347-50), **malaria** (*Anopheles* mosquito, protozoan parasite), yellow fever (*Aedes*, virus). Before synthetic pesticides: **pyrethrum** (pyrethrins, chrysanthemum flowers), **nicotine sulfate** (tobacco), garlic oil, companion planting.

The dirty dozen (Stockholm Convention, 2001) ★

POP	Use
Aldrin	crop insecticide (corn, cotton)
Chlordane	crop insecticide (vegetables, citrus, cotton, potatoes)
DDT	insecticide (cotton)
Dieldrin	crop insecticide (cotton, corn)
Endrin	crop insecticide (cotton, grains)
Heptachlor	insecticide (termites, soil insects)
Hexachlorobenzene (HCB)	fungicide for seed treatment
Mirex	insecticide (termites, fire ants)
Toxaphene	insecticide (livestock and **cotton**) ★; $C_{10}H_{10}Cl_8$ ★
PCBs	industrial: heat-exchange fluid in transformers, paints, plastic additive
Dioxins	unintentionally produced during combustion
Furans	unintentionally produced during combustion

- 10 of the 12 are made intentionally; 9 are insecticides/fungicides; dioxins and furans are unintentional by-products. Later additions: **8 more added in 2009** (incl. 5 more polychlorinated insecticides, polybrominated flame retardants, PFOS from Scotchgard) and **1 more in 2011** (slides 34, 39).

Slides from the course deck (Kishore Class 1-3)

Organic Chemicals in the Environment

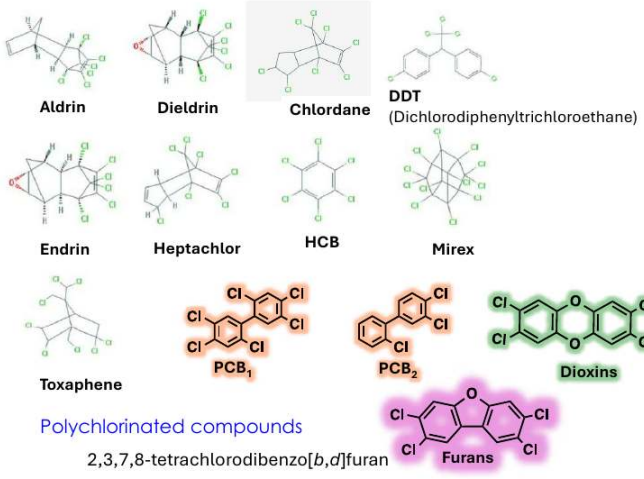
The 12 Key POPs—the Dirty Dozen

POP	Use
Aldrin	crop insecticide (corn, cotton)
Chlordane	crop insecticide (vegetables, citrus, cotton, potatoes)
DDT	crop insecticide (cotton)
Dieldrin	crop insecticide (cotton, corn)
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Heptachlor	insecticide (termites and soil insects)
Hexa-chlorobenzene	fungicide for seed treatment
Mirex	insecticide (termites, fire ants)
Toxaphene	insecticide (livestock and crops)
PCBs	industrial chemical (heat exchange fluid for electrical transformers, paint and plastic additive)
Dioxins	unintentionally produced during combustion
Furans	unintentionally produced during combustion

27

Kishore Class 1-3 · slide 27

Organic Chemicals in the Environment



Polychlorinated compounds

2,3,7,8-tetrachlorodibenzo[*b*,*d*]furan

28

Kishore Class 1-3 · slide 28

Organic Chemicals in the Environment

Stockholm Convention

Added 8 more substances to the agreement in 2009 and one more in 2011

New POPs Added to the Stockholm Convention in 2009 and 2011

POP	Use
Hexachlorocyclohexane	Unintentional by-product of lindane production
Chlordecone	Insecticide (ant and roach)
Hexabromobiphenyl	Fire-suppressing chemical
Hexabromodiphenyl and modiphenyl ether / Heptabro-	Flame retardant
Lindane	Broad-spectrum insecticide
Pentachlorobenzene	PCB
Perfluorooctane sulfonic acid	Key ingredient in Scotchgard fabric protector
Tetrabromodiphenyl ether	Flame retardant
Endosulfan	Crop insecticide

34

Kishore Class 1-3 · slide 34

Organic Chemicals in the Environment

Stockholm Convention

Added 8 more substances to the agreement in 2009 and one more in 2011

- Five of the new POPs are also poly chlorinated compounds which are used as insecticides
- Polychlorinated hydrocarbons are insoluble in water but soluble in fats
- Polychlorinated hydrocarbons concentrated in the fatty tissues of fish and of birds, and of humans who eat them
- For the first time, polybrominated compounds, used in the 1970s as flame-retardant agents, were added to the list.
- Perfluorooctane sulfonic acid, the key ingredient in Scotchgard fabric protector, was added to the list of new POPs



39

Kishore Class 1-3 · slide 39

DDT ★

- First made 1874 (Zeidler); insecticidal property 1939 (**Müller; Nobel 1948**). Cheap, persistent, low acute mammalian toxicity; used for malaria and typhus control.
- Problems: resistance; accumulation in fatty tissue; fish-eating birds (bald eagle, peregrine falcon, brown pelican) lay thin-shelled eggs because DDE interferes with **calcium metabolism** in eggshell formation. Water-insoluble → sediment → plankton → small fish → large fish → raptors: **biomagnification**.
- Rachel Carson (*Silent Spring*) → EPA created (1970) → **DDT banned in the US in 1973**. In India agricultural use banned, public-health (malaria vector control) use continued. Environmental scientist vs environmental activist slide (57).

Dioxins (2,3,7,8-TCDD) and furans

- Formed unintentionally by combustion of chlorinated waste, chlorine bleaching of pulp, and as an impurity in trichlorophenol manufacture. **TCDD** = 2,3,7,8-tetrachlorodibenzo-*p*-dioxin, extremely toxic; persistent, bioaccumulative, carcinogenic. Impurity in **Agent Orange** (Vietnam defoliant, 2,4-D/2,4,5-T). Furan example: 2,3,7,8-tetrachlorodibenzofuran.

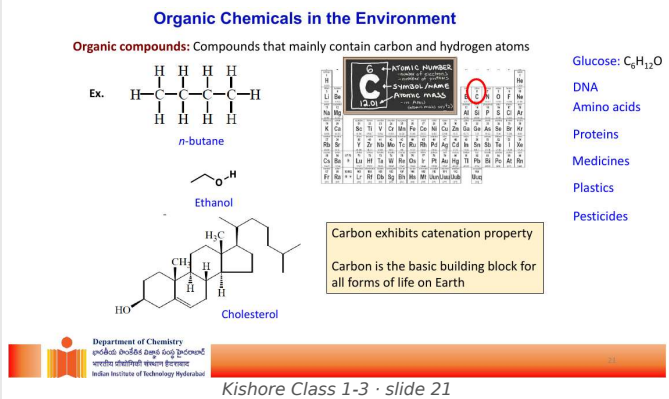
PCBs ★

- **Polychlorinated biphenyls** ★ (2024 Q41): biphenyl with 1-10 Cl; chemically stable, non-flammable, non-conducting, water-insoluble, fat-soluble → transformer and capacitor fluids, plasticisers. Concentrations in fish can be >**1 million×** that in water. Health: chloracne, liver damage, reproductive and developmental effects, cancer; birds: eggshell thinning. **US production halted 1977**; still made elsewhere (e.g. Russia). Coplanar PCBs are dioxin-like.

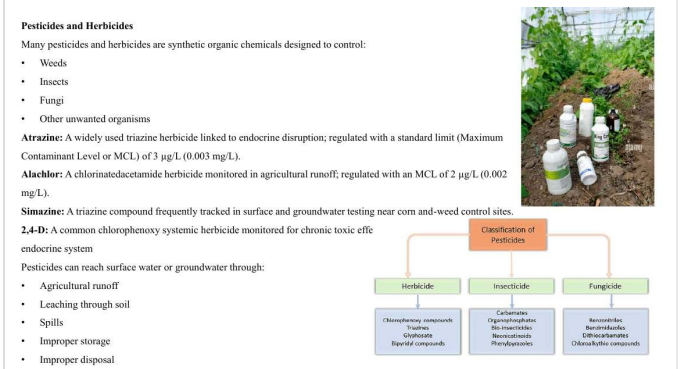
Other synthetic organic chemicals in water

- Pathways to drinking water: agricultural runoff and leaching, industrial discharge, landfill leachate, wastewater, atmospheric deposition. Health effects vary, from nervous-system, liver, kidney, reproductive and cancer risks to endocrine disruption.
- Pesticides/herbicides: **atrazine** (triazine; MCL 3 µg/L), **alachlor** (chloroacetamide; MCL 2 µg/L), simazine, 2,4-D (chlorophenoxy).
- VOCs and industrial solvents: benzene, toluene, **TCE** (trichloroethylene; degreaser) ★, **PCE** (perchloroethylene; dry-cleaning). **PFAS** (PFOA, PFOS): fluorinated; water/oil/grease resistant; firefighting foams, non-stick; extremely persistent ('forever chemicals'). Phthalates and BPA are endocrine disruptors.
- Classification of organic pollutants in agricultural environments (slide 48-49): pesticides, antibiotics and veterinary drugs, plasticisers/film residues, PAHs, PCBs; spread through soil, water, air and crops.

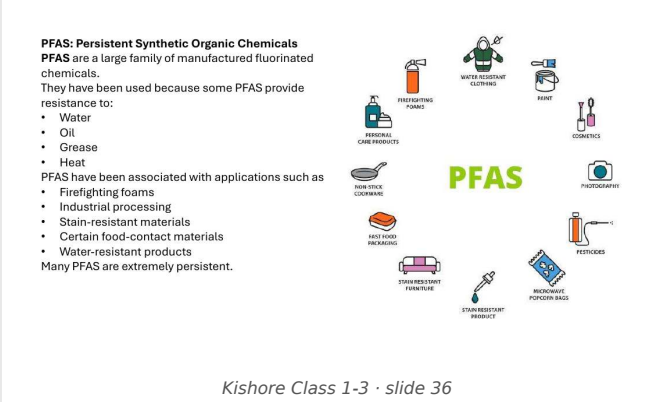
Slides from the course deck (Kishore Class 1-3)



Kishore Class 1-3 · slide 21



Kishore Class 1-3 · slide 33



Kishore Class 1-3 · slide 36



Kishore Class 1-3 · slide 41

Organic Chemicals in the Environment

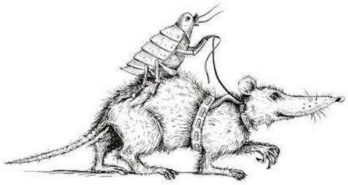
Why are we producing dangerous synthetic chemicals intentionally?

- Human civilization always have had to contend with pests

- **The three most impactful pests in human history:**

1) Rats and fleas, which spread the devastating Black Death (1347-1350)

Black Death: an infectious fever caused by the **Bacterium Yersinia pestis**. The disease was likely transmitted from **rodents** to humans by the bite of infected **fleas**





Kishore Class 1-3 · slide 42

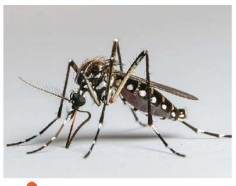
Organic Chemicals in the Environment

Why are we producing these dangerous chemicals intentionally?

- Human civilization always have had to contend with pests

- **The three most impactful pests in human history:**

1) Rats and fleas, which spread the devastating Black Death (1347-1350);
2) Mosquitoes, which carry diseases like malaria and yellow fever;
3) Viruses, responsible for smallpox and Spanish flu





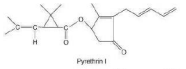
Kishore Class 1-3 · slide 43

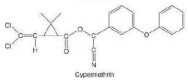
Organic Chemicals in the Environment

What were we using before Synthetic Pesticides came into use?

- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

Examples:
1. Pyrethrins obtained from the pyrethrum flower







Kishore Class 1-3 · slide 44

Organic Chemicals in the Environment

Companion Plantation of pyrethrum plants in **organic farming**



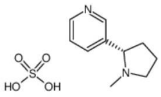
Kishore Class 1-3 · slide 45

Organic Chemicals in the Environment

What were we using before Synthetic Pesticides came into use?

- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

Examples:
2. Nicotine sulfate; Nicotine obtained from tobacco plants is neutralized with sulfuric acid





Kishore Class 1-3 · slide 46

Organic Chemicals in the Environment

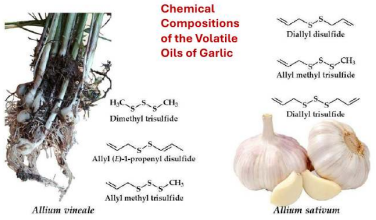
What were we using before Synthetic Pesticides came into use?

- Before 1940, only a few pesticides were available and most of the insect poisons naturally extracted from plants

Examples:
3) Garlic oil

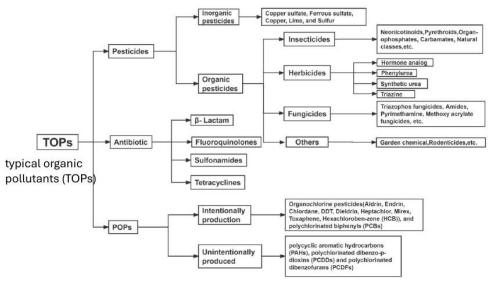


Chemical Compositions of the Volatile Oils of Garlic



Kishore Class 1-3 · slide 47

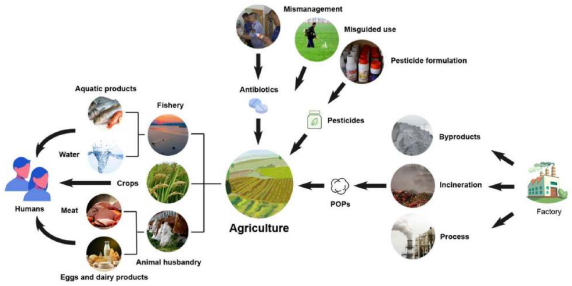
The classification of typical organic pollutants in the agricultural environment



https://doi.org/10.3390/agronomy12123009

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The sources and spread of typical organic pollutants in the agricultural environment.

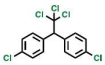


https://doi.org/10.3390/agronomy12123009

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Slides from the course deck (Kishore Class 1-3)

Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)



- The introduction of DDT at the end of WW-II (late 1940s) escalated the use of pesticides and the development of several other pesticides

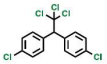
Advantages

- Inexpensive
- Nontoxic to humans and other mammals
- Stable
- Continued to kill insects for a long period after application

Discovery and History
Created in 1874: Austrian chemist Othmar Zeidler first made the chemical. Found useful in 1939: Swiss chemist Paul Hermann Müller discovered that it killed insects. He won a Nobel Prize for this in 1948.

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Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)





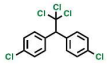


Disadvantages

- Harmful to fish-eating birds such as bald eagles, peregrine falcons, and brown pelicans
- Water gets contaminated with DDT from aerial spraying and runoff from treated land
- With time, insects also became increasingly resistant to DDT and larger amounts had to be applied to achieve the desired results
- DDT interferes with calcium metabolism in birds and it causes eggshells to become thin and break when parent birds attempt to incubate the eggs

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Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)





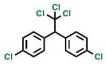


How does DDT get into food chain if it is insoluble in water?

- Contaminated runoff seeps into water bodies, then since the DDT is insoluble in water, it settles in the sediments.
- Plankton absorb it from the sediments in water.
- This DDT is then transferred to small fish when they eat the plankton, and the concentration increases further as larger fish consume the small fish, leading to biomagnification.
- The highest concentrations of DDT accumulate in the top predators, such as fish-eating birds or humans.

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Synthetic Organic Chemicals in the Environment



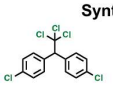
DDT
(Dichlorodiphenyltrichloroethane)



- The first person to draw public attention to the dangers of pesticides was biologist **Rachel Carson** and she played a key role in the establishment of Environmental Protection Agency (EPA) in 1970
- In 1973, the use of DDT was banned in the United States and fish-eating birds population restored shortly
- Still DDT is partially banned in several countries such as India
- DDT is banned for agriculture but allowed for fumigation to kill mosquitoes to prevent malaria

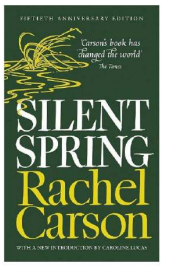
Kishore Class 1-3 · slide 55

Synthetic Organic Chemicals in the Environment



DDT
(Dichlorodiphenyltrichloroethane)

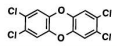




Kishore Class 1-3 · slide 56

Synthetic Organic Chemicals in the Environment

Dioxins 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD)



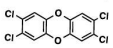
Produced by burning chlorine containing medical and municipal wastes

- TCDD persists in the environment for a very long time and bioaccumulates within the food chain
- TCDD is extraordinarily toxic to guinea pigs and is a known carcinogen for many animals
- During Vietnam war, USA used **Agent Orange** (defoliant) in which TCDD present as an impurity
- EPA concluded that exposure to dioxin, increase the risk of cancer, may disrupt some reproductive mechanisms and suppress the immune system.

Kishore Class 1-3 · slide 58

Synthetic Organic Chemicals in the Environment

Dioxins 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD)



Major sources for TCDD

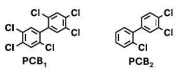
- Burning of chlorine-containing **medical and municipal** wastes
- **Paper mills** that use chlorine to bleach paper pulp
- During the manufacture of **trichlorophenol** a precursor for herbicides

Currently, proper solutions were found for above three problems

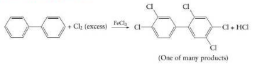
Kishore Class 1-3 · slide 59

Synthetic Organic Chemicals in the Environment

PCBs (Polychlorobiphenyls)



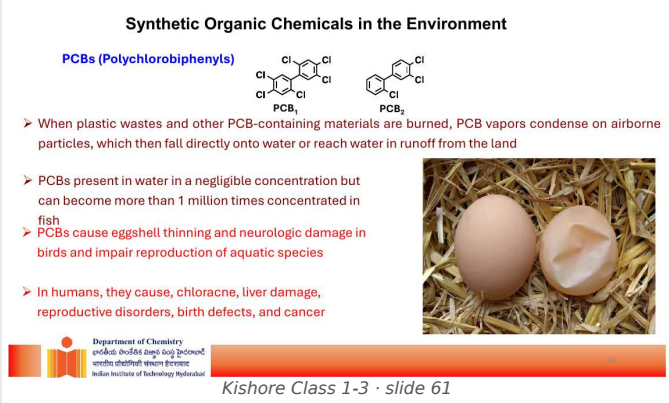
- PCBs are not water soluble but readily soluble in fat
- PCBs are chemically and thermally stable, fire and electricity resistant and persist in the environment for many years



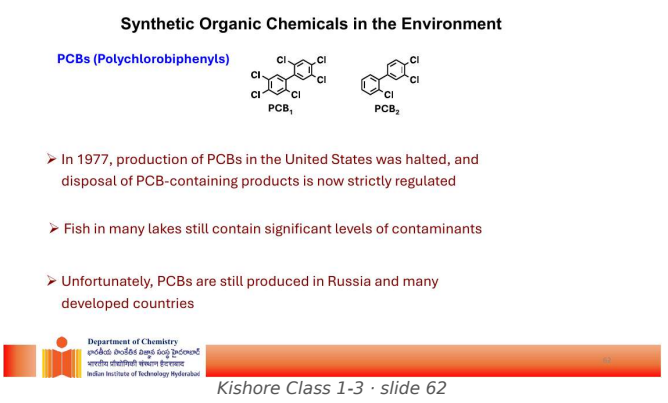
Uses of PCBs

- Heat exchange fluid in transformers,
- Paints and plastic additive (plasticizer)

Kishore Class 1-3 · slide 60



Kishore Class 1-3 · slide 61



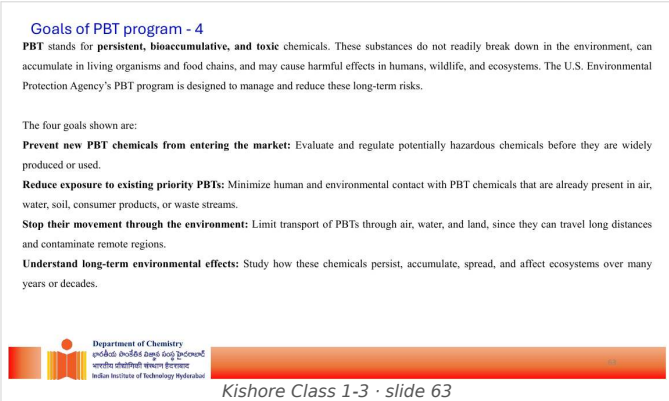
Kishore Class 1-3 · slide 62

B3. Environmental fate, K_{ow}, PBT and TSCA (Class 1-3)

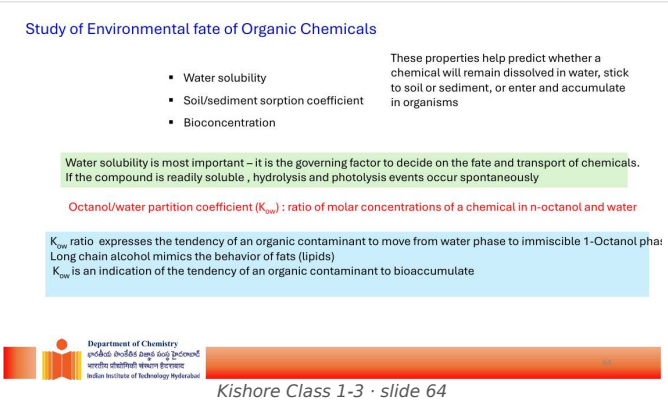
- Fate depends on **water solubility** (controls transport and fate), sorption to soil/sediment, volatility, degradation and bioconcentration. **PBT** = persistent, bioaccumulative, toxic. EPA PBT program has 4 goals: prevent new PBTs, reduce exposure to existing ones, prevent movement and transfer, and understand long-term effects.
- **K_{ow} = [chemical]$_{octanol}$ / [chemical]$_{water}$**; octanol mimics fat. **High K_{ow} (hydrophobic) → bioaccumulates, adsorbs to sediment, low water solubility.** log K_{ow} rises with the number of chlorine atoms and the number of aromatic rings (linear plots on slide 65). Lowest K_{ow} = most water-soluble, mobile and degradable.
- PBT profiler answers three questions for manufacturers: where will the chemical end up (air/water/soil/sediment), how long will it last (persistence), how toxic is it (hazard).
- **TSCA** (Toxic Substances Control Act): chemical makers file **premanufacture notifications (PMNs)**; EPA may restrict or ban a chemical that poses unreasonable risk.

Compound (slide 66)	log K_{ow}	Compound	log K_{ow}
aldrin	6.5	dieldrin	4.3
DDT	6.2	1,2,4-trichlorobenzene	4.2
hexachlorobenzene	5.6	dibutyl phthalate	4.1
chlordane	5.5	dibenzyl	4.0
heptachlor	5.3	ortho-dichlorobenzene	3.7
toxaphene	4.8	1,4-dichlorobenzene	3.6
fluoranthene	4.7	chlorobenzene	2.8
endrin	4.6	toluene	2.7
phenanthrene	4.5	benzene / phenol	2.1 / 1.5

Slides from the course deck (Kishore Class 1-3)

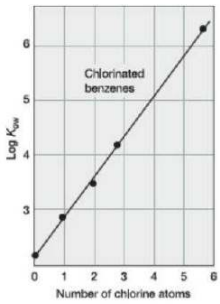


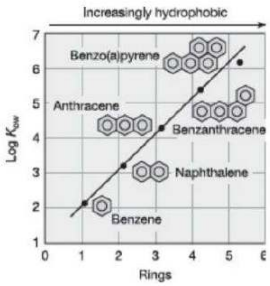
Kishore Class 1-3 · slide 63



Kishore Class 1-3 · slide 64

Environmental Assessment based on K_{ow}





Department of Chemistry
భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Kishore Class 1-3 · slide 65

Which one has a more tendency to degrade?

Compound	Log K_{ow}
aldrin	6.5
DDT	6.2
hexachlorobenzene	5.6
chlordane	5.5
heptachlor	5.3
toxaphene	4.8
fluoranthene	4.7
Endrin	4.6
phenanthrene	4.5
Dieldrin	4.3
1,2,4-trichlorobenzene	4.2
dibutyl phthalate	4.1
dibenzyl	4.0
1,4-dichlorobenzene	3.6
ortho-dichlorobenzene	3.7
chlorobenzene	2.8
toluene	2.7
benzene	2.1
phenol	1.5

Department of Chemistry
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Indian Institute of Technology Hyderabad

Kishore Class 1-3 · slide 66

PBT profile is intended to answer three questions for manufacturers?

1. Once released, will a chemical go into the air, water, soil, or sediment?
2. How long will this chemical stay in the medium (air, water, soil or sediment)?
3. Will the chemical present hazard?

PBT profiler calculates an atmospheric half-life by determining the importance of a chemical's reaction with two of the most prevalent atmospheric oxidants, **hydroxyl radicals** and **ozone**.

The half-life is calculated directly from gas phase hydroxyl radical and ozone reaction rate constants.

The 12 EPA's Persistence Criteria for PBT Chemicals

	Half-Life of Chemical		
Media	Not Persistent	Persistent	Highly Persistent
Water	< 2 months	≥ 2 months	> 6 months
Soil	< 2 months	≥ 2 months	> 6 months
Air	≤ 2 days		> 2 days
Sediment	< 2 months	≥ 2 months	> 6 months

Department of Chemistry
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Kishore Class 1-3 · slide 67

Under **Toxic Substances Control Act (TSCA)**, chemical manufacturers are required by law to submit **premanufacture notifications (PMNs)** to the EPA for new chemicals that they are planning to manufacture domestically or import on an industrial scale.

The EPA uses a sophisticated computer program that evaluates a chemical's characteristics to ascertain whether it might pose an unreasonable risk to human health or the environment.

Based on this evaluation, the EPA can stop the production of these chemicals until the manufacturer can prove that they will not pose an unreasonable risk if released into the environment.

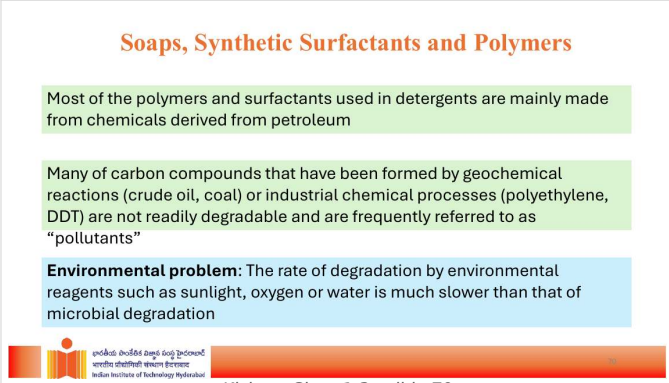
Department of Chemistry
భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Kishore Class 1-3 · slide 68

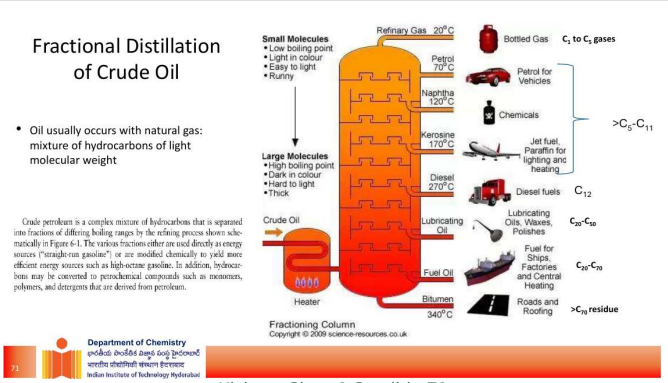
B4. Soaps, synthetic surfactants and detergents (Class 1-3)

- Most polymers and surfactants come from petroleum. Fractional distillation: C_1-C_5 gases; C_5-C_{11} gasoline; ~C_{12} kerosene; C_{20}-C_{50} fuel/lube oils; C_{20}-C_{70} waxes and lubricants; >C_{70} asphalt. Degradation of these compounds by sunlight/O_2/water (abiotic) is much slower than microbial degradation.
- **Soap** = Na (or K) salt of long-chain fatty acids made by **saponification**: fat/oil (triglyceride) + NaOH → glycerol + fatty acid salts; **salting out with NaCl**. Stearate (C18 saturated) from animal fat; oleate (C18:1) from olive oil; palm oil gives oleate and palmitate (C16) in about 1:1.
- Micelle: hydrophobic tails inward, hydrophilic head groups outward; surfactant types: anionic, cationic, non-ionic, amphoteric. Fatty acids are made from acetate and are broken back to acetate by microbes (linear chains → readily biodegradable).
- Soap drawbacks: reacts with Ca^{2+}/Mg^{2+} in hard water to precipitate scum; protonates in acid (loses solubility).
- **Synthetic surfactants**: alkyl sulfonates, alcohol sulfates, ether sulfates etc.; no hard-water precipitates. Early branched-alkylbenzene sulfonates persisted (foaming); linear alkylbenzene sulfonates (LAS) biodegrade.
- **Detergent** = surfactant + **builders** + bleach + enzymes + brighteners. Builders inactivate Ca^{2+}/Mg^{2+}: polyphosphates (sodium tripolyphosphate), zeolites, silicates, polycarboxylates. Polyphosphates → eutrophication/algal blooms → replaced by zeolites; yet detergents are only **20-25%** of phosphate in lakes (rest: fertilisers, animal waste, sewage).
- Seashells and calcium carbonate deposits (slide 81) illustrate accumulation of natural carbonates over centuries.

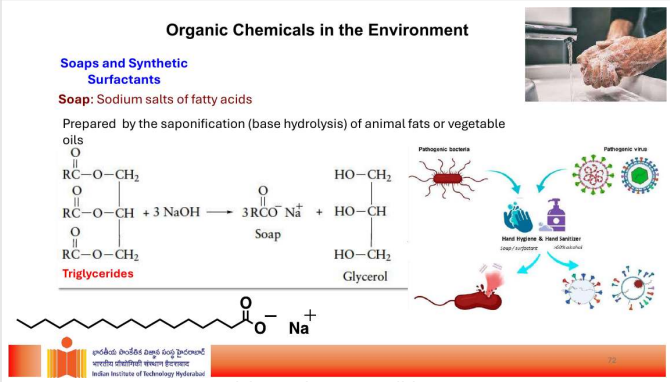
Slides from the course deck (Kishore Class 1-3)



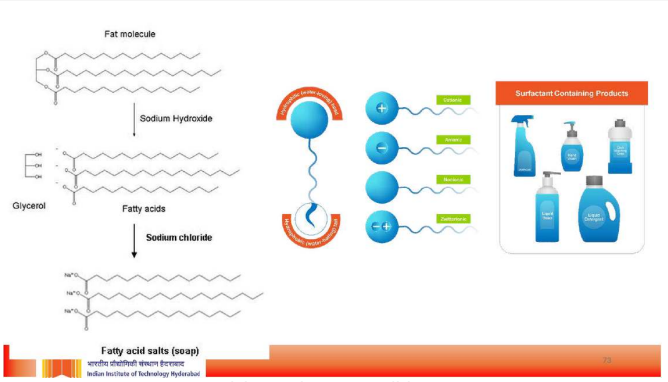
Kishore Class 1-3 · slide 70



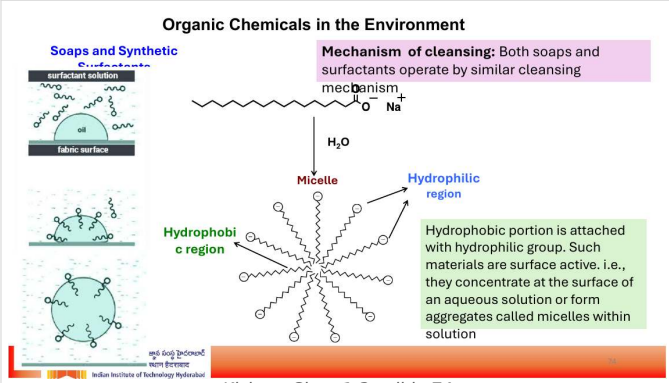
Kishore Class 1-3 · slide 71



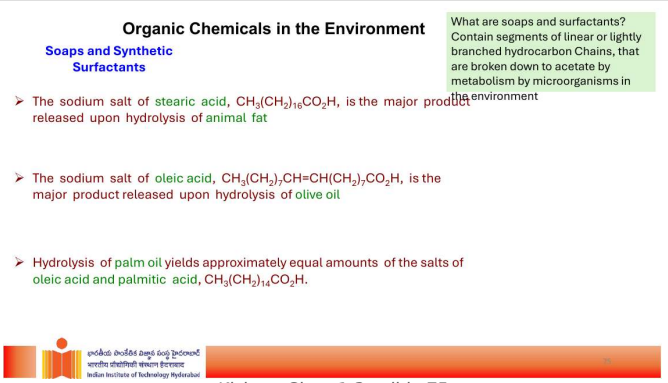
Kishore Class 1-3 · slide 72



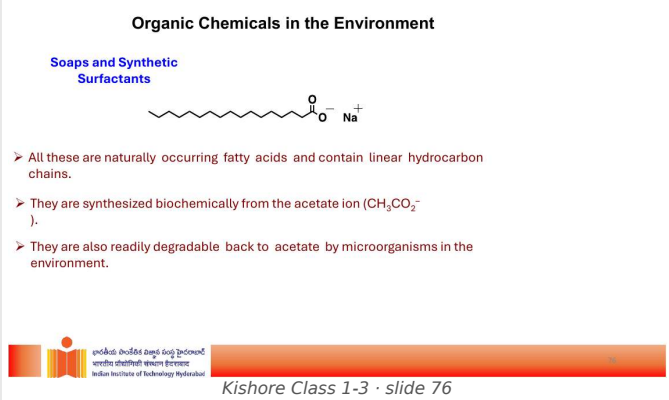
Kishore Class 1-3 · slide 73



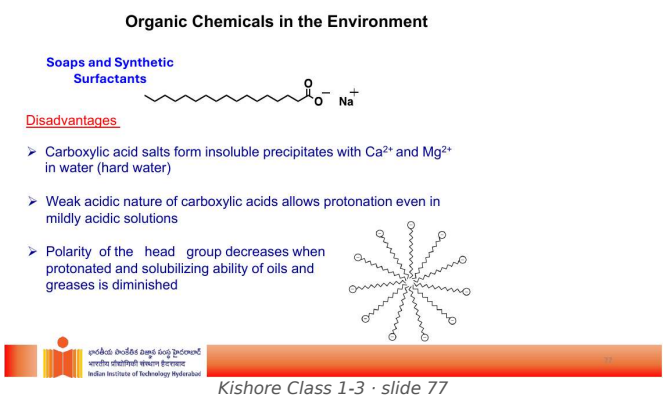
Kishore Class 1-3 · slide 74



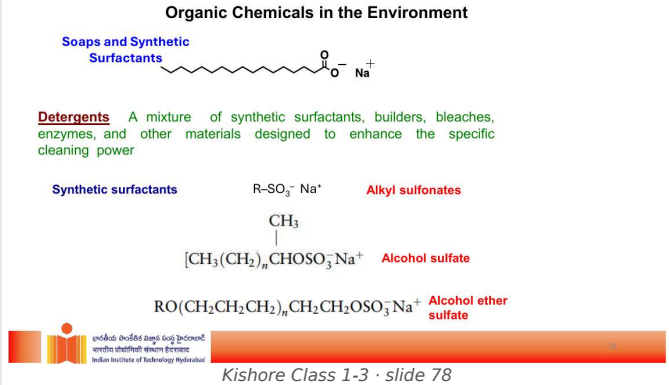
Kishore Class 1-3 · slide 75



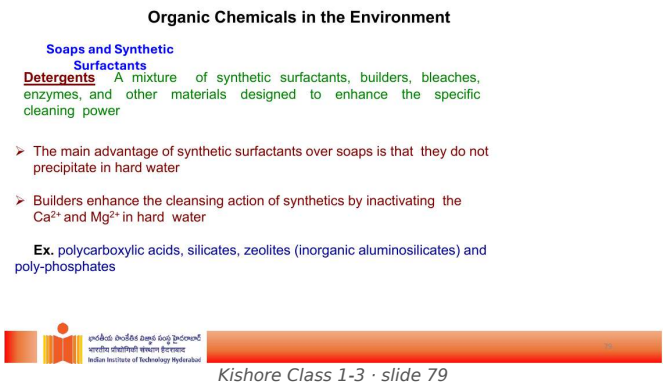
Kishore Class 1-3 · slide 76



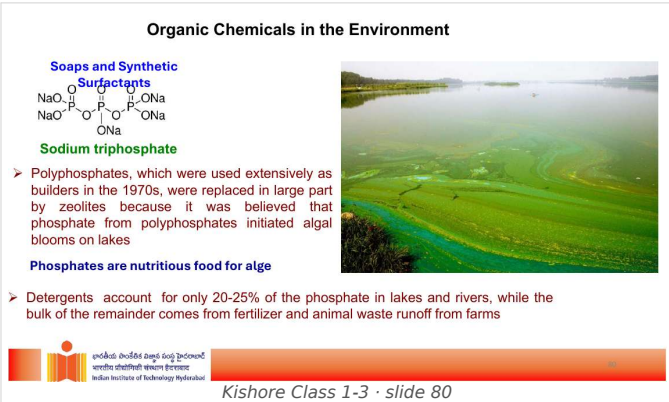
Kishore Class 1-3 · slide 77



Kishore Class 1-3 · slide 78



Kishore Class 1-3 · slide 79



Kishore Class 1-3 · slide 80



Kishore Class 1-3 · slide 81

B5. Polymers (Class 1-3)

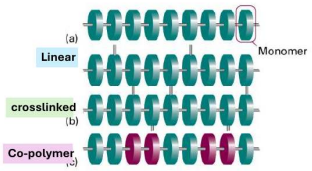
- **Macromolecules** are very large molecules built from small repeating units; polymers: n repeat units; mass ~10-100 kDa. Natural: cellulose, starch (sugars), **proteins (amino acids)** ★, DNA/RNA (nucleotides), natural rubber, wool. Synthetic: PE, PP, PS, PVC, PET, nylon, Teflon. 'Which are polymers?' → **all** (slide 83).
- Structures: linear, branched, cross-linked; co-polymers. Natural polymers are generally biodegradable; **natural rubber** (10,000-20,000 isoprene units) is not easily biodegradable and is water-resistant.
- Uses: packaging films/bags (PE), ropes/tapes (PP), foams (PS), lenses (PMMA), pipes/frames (PVC), air bags/carpets/ropes (nylon), water bottles (PET).
- **Chain-growth (addition) polymerisation**: initiator ROOR → 2RO•; radical adds to monomer (acrylonitrile → **polyacrylonitrile**); termination by combination or disproportionation.
- **Step-growth (condensation)**: adipic acid + hexamethylenediamine → **nylon** ($-H_2O$); ethylene glycol + terephthalic acid → **PET**. Monomers react to oligomers, which then react with each other.
- **Additives**: plasticisers soften polymers (PVC); **DEHP** = di(2-ethylhexyl) phthalate; di-n-butyl phthalate; they leach and are endocrine disruptors; use restricted/banned (slide: after 2000).
- Rate of decomposition ★ (2024 Q40): sugar/starch and protein degrade faster; synthetic plastics are the slowest-decomposing.

Slides from the course deck (Kishore Class 1-3)

Macromolecules

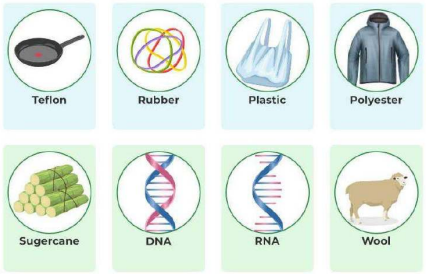
Macromolecules are very large molecules assembled from smaller molecules
(i) biosynthetically in organisms,
(ii) by chemists in the laboratory, or in an industrial reactor.

Naturally occurring macromolecules include
(i) polysaccharides such as cellulose,
(ii) polypeptides such as protein enzymes, and
(iii) polynucleotides such as deoxyribonucleic acid (DNA).



Kishore Class 1-3 · slide 82

Which of them are polymers?



ALL

Kishore Class 1-3 · slide 83

Organic Chemicals in the Environment

Polymers

- Polymers are also called macromolecules
- They have very large size and high molecular weight (10 kDa to 100 kDa)

Ex. $-[CH_2CH_2]-_n$

Polyethylene

n = degree of polymerization

Synthesized from ethylene $H_2C=CH_2$

Kishore Class 1-3 · slide 84

Organic Chemicals in the Environment

Polymers

Polymer	Repeat unit	Monomer(s)	
Polyethylene	$-[CH_2CH_2]-_n$	$H_2C=CH_2$	Packaging films, bags, bottles and toys
Polypropylene	$-[CHCH_2]-_n$ (CH_3)	$CH_3HC=CH_2$	Ropes, tapes, carpets
Polystyrene	$-[CHCH_2]-_n$ (C$_6$H$_5$)	$CH=CH_2$ (C$_6$H$_5$)	Packaging foam, Sterilizing test tubes, lids, bottles
Poly(methyl methacrylate)	$-[CH_2C(CH_3)(COOCH_3)]-_n$	$H_2C=C(CH_3)COOCH_3$	Artificial teeth, Bathtubs, Contact lenses
Poly(vinyl chloride)	$-[CH_2CHCl]-_n$	$H_2C=CHCl$	Window frames, pipes, medical devices



Kishore Class 1-3 · slide 85

Organic Chemicals in the Environment

Polymer	Repeat unit	Monomer(s)	
Nylon 66	$-[C(=O)(CH_2)_4C(=O)NH(CH_2)_6NH]-_n$	$HOC(=O)(CH_2)_4COH$ and $NH_2(CH_2)_6NH_2$	Air bags, carpets, ropes
Poly(ethylene terephthalate) (PET)	$-[C(=O)C_6H_4C(=O)OCH_2CH_2O]-_n$	$HOCH_2CH_2OH$ and $HOC(=O)C_6H_4C(=O)OH$	Water bottles, Containers, Bags





Kishore Class 1-3 · slide 86

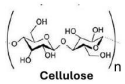
Organic Chemicals in the Environment

Polymers

- Many macromolecules are components of living systems

Proteins: polymers of amino acids

Starch and Cellulose: polymers of sugars



Cellulose

These naturally occurring polymers are biodegradable and will not cause long-term environmental problems

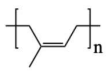
Kishore Class 1-3 · slide 87

Organic Chemicals in the Environment

Polymers

- Natural rubber is not easily biodegradable

In natural rubber 10,000 to 20,000 isoprene units are linked together.

Major component 

cis-1,4- polyisoprene



Highly resistant to water

$n\,CH_2=C(CH_3)-CH=CH_2 \xrightarrow{\text{Polymerisation}} -[CH_2-C(CH_3)=CH-CH_2]-_n$

Isoprene
2-methyl–1, 3–butadiene

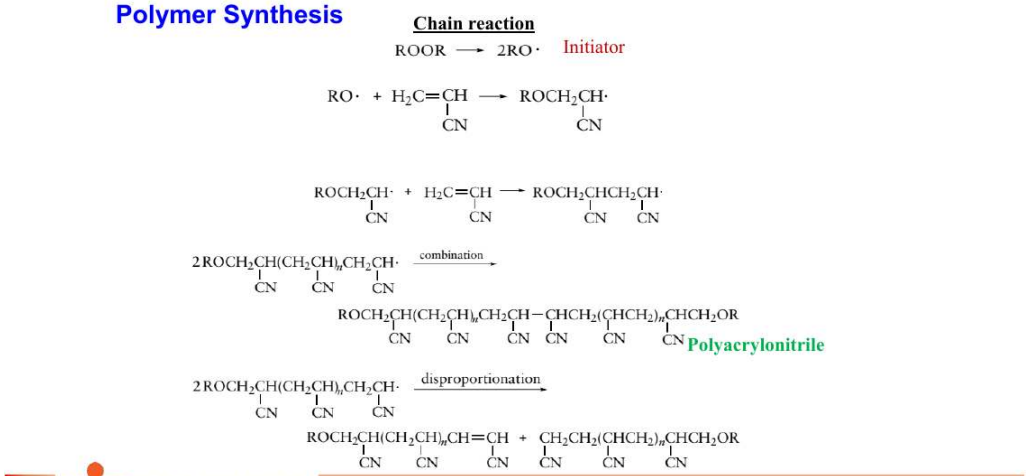
Polyisoprene
(natural rubber)

Kishore Class 1-3 · slide 88

Organic Chemicals in the Environment

Polymer Synthesis

Chain reaction

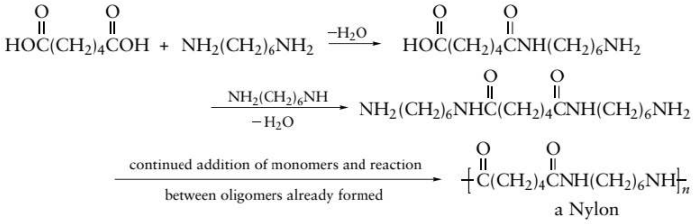


Kishore Class 1-3 · slide 89

Organic Chemicals in the Environment

Polymer Synthesis

Step reaction

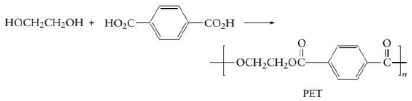


Kishore Class 1-3 · slide 90

Organic Chemicals in the Environment

Polymer Synthesis

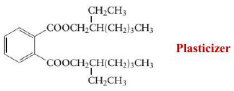
Step reaction



Kishore Class 1-3 · slide 91

Organic Chemicals in the Environment

Polymer Additives



- DHEP commonly used as a plasticizer in vinyl polymers
- DHEP can leach from vinyl polymers and cause serious medical problems
- After 2000, use of DHEP as a plasticizer is banned

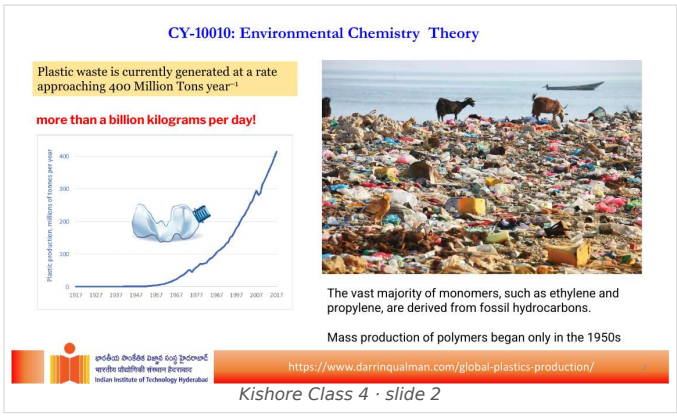
Plasticizers : Certain plastics do not soften much on heating. These can be easily softened by the addition of some organic compounds which are called plasticizers. For example, polyvinyl chloride (PVC) is very stiff and hard but is made soft by adding di-n-butylphthalate (a plasticizer). Some other common plasticizers are dialkyl phthalates and cresyl phthalate

Kishore Class 1-3 · slide 92

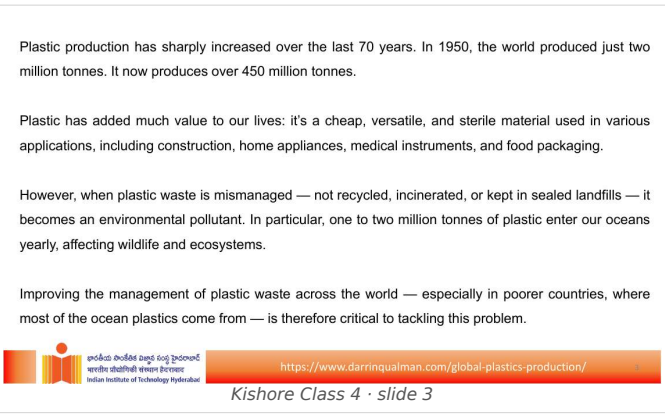
B6. Plastic waste and its fate (Class 4)

- Plastic production rose from **2 Mt in 1950** to >**450 Mt** today; waste is ~**400 Mt/yr**. Monomers (ethylene, propylene) come from fossil hydrocarbons. Mass production since the 1950s; durability is the problem (strong covalent bonds).
- Ocean input: ⚠ Class 4 says ~1-2 Mt per year; Class 1-3 says ~8 Mt. Decomposition times: food 2-10 d; paper 2-5 months; leather 25-40 y; plastic bags 15-1000 y; bottle >1000 y (⚠ Class 1-3: 20-500 y).
- **Geyer et al.** fate of all plastic ever made: **18% recycled, 24% incinerated (energy), 58% landfill or environment**; projected ~**12,000 Mt** of plastic waste accumulated by 2050. Impact: seabirds, seals, turtles, whales; entanglement; microplastics.
- Degradation rate r_d (polymer degradation slides 10-12): depends on chain length, crystallinity, additives, sunlight, temperature, microbes.
- **Waste hierarchy** (best → worst): **source reduction → reuse and recycle → compost → volume reduction (processing, WtE) → landfill**. Source reduction avoids generation; reuse extends product life; recycling examples (mobile sets, old cars); compost = nature's recycling; processing reduces volume.
- **Waste-to-energy** (Vienna incinerator): 7 stages: pit/feeding → combustion chamber → boiler (steam) → turbine/generator → air-pollution control → ash handling → stack. **Waste-to-wealth**: depolymerisation; **pyrolysis** (thermal decomposition) with catalysts such as zeolites/cobalt → fuels; chemical upcycling of **polystyrene** and **PVC** (slides 30-32).
- Landfills today are lined and engineered, the last resort. Waste minimisation by process manipulation: industries purify and recycle process streams (e.g. HCl recovery); waste exchanges.
- Hazardous-waste policy: (1) minimise generation, (2) convert to less hazardous forms, (3) store securely. Protect Earth: conserve water and energy, plant trees, reduce single-use plastic ('RRR').

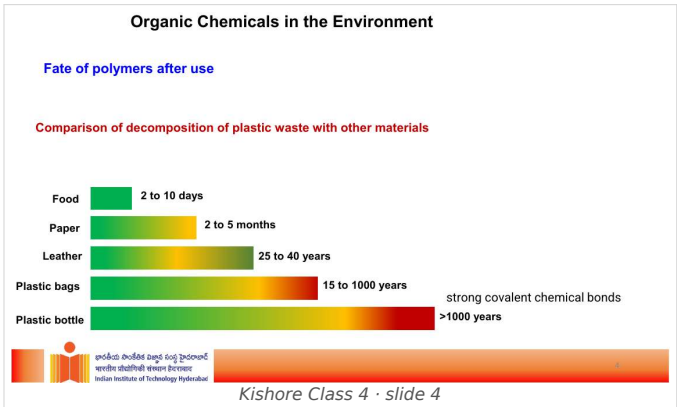
Slides from the course deck (Kishore Class 4)



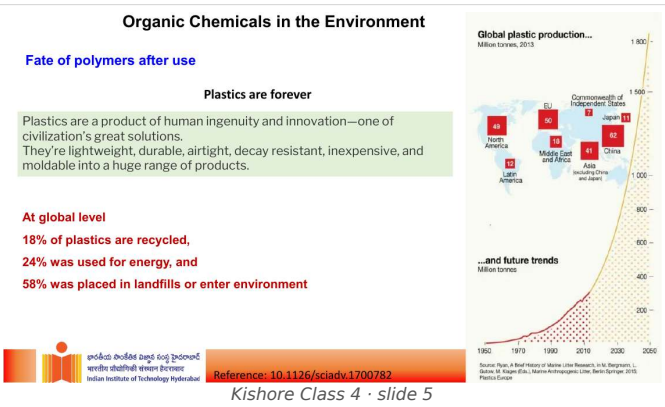
Kishore Class 4 · slide 2



Kishore Class 4 · slide 3



Kishore Class 4 · slide 4

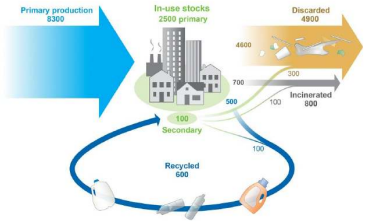


Kishore Class 4 · slide 5

Organic Chemicals in the Environment

Fate of polymers after use

Global production, use, and fate of polymers (1950 to 2025 in million metric tons)



Reference: 10.1126/sciadv.1700782

Kishore Class 4 · slide 6

Uses Of Plastic In Building Construction

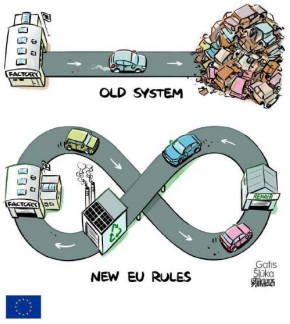


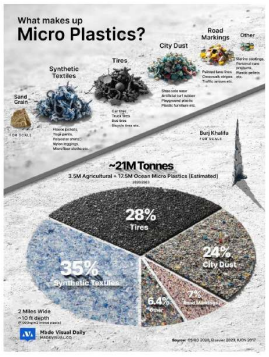
Kishore Class 4 · slide 7





Kishore Class 4 · slide 8





Kishore Class 4 · slide 9

Degradation rates of plastics

The polymer degradation rate, r_d, is the differential mass loss per unit time

$$r_d = -\frac{dm}{dt} = K \cdot SA$$

Table 1. Estimated Half-Lives of Common Plastic Items[a]

Plastics type	Common applications	Typical thickness (μm)	Estimated Specific Surface Degradation Rate (min−max; μm year⁻¹)[b] Land (buried)	Land (accel. by UV/heat)[c]	Marine	Marine (accel. by UV/heat)[c]
1. PET	Single-use water bottle	500	0	–	–	110
2. HDPE	Plastic bottles	500	1.0 (0.91–1.1)	1.3 (0.55–2.6)	4.3 (0–11)	9.5 (4.5–22)
2. HDPE	Pipes	10,000	1.0 (0.91–1.1)	1.3 (0.55–2.6)	4.3 (0–11)	9.5 (4.5–22)
3. PVC	Pipes	10,000	0	–	–	–
4. LDPE	Plastic bags	100	11	22 (1.6–83)	15 (0–37)	10 (9.0–12)
5. PP	Food storage container	800	–	0.51	7.5	4.6
6. PS	Insulating packaging	20,000	0	–	–	–
7. Others	Biodegradable plastic bag	100	270 (20–1400)	320	16 (7.5–29)	180

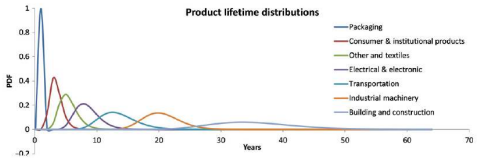
https://www.epa.gov
https://pubs.acs.org/doi/epdf/10.1021/acssuschemeng.9b06635

Kishore Class 4 · slide 10

Organic Chemicals in the Environment

Fate of polymers after use

How long plastics are in use before they reach the end of their useful lifetimes



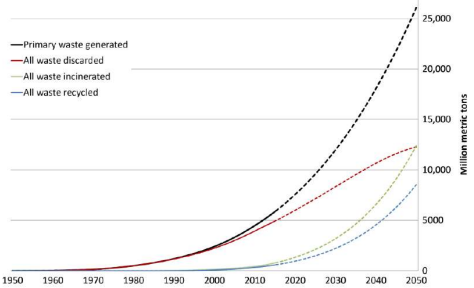
At current growth rates, the accumulation of plastics waste in landfills and/or in the natural environment is projected to reach nearly 12,000 Million tons globally by 2050

Reference: 10.1126/sciadv.1700782

Kishore Class 4 · slide 11

Fate of polymers after use

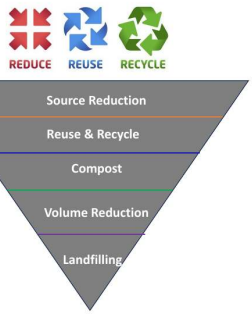
Cumulative plastic waste generation and disposal



Reference: https://dx.doi.org/10.1021/acssuschemeng.9b06635?ref=pdf

Kishore Class 4 · slide 12

Waste Hierarchy



Kishore Class 4 · slide 13

Recycle

- Every day we use products made from recycled materials.
- Take your glass, cans, newspapers, milk jugs and other acceptable recyclable items to your local transfer station, drop off location or place out for curbside collection so that they can be turned into new products like jackets
- Frisbees, paper products, and soda cans. Recycling saves money, energy, and the environment.
- 48% of municipal solid waste was recycled in 2015.

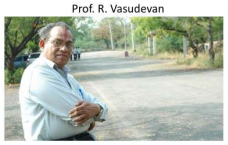




Kishore Class 4 · slide 16

Solution to plastic pollution

Prof. R. Vasudevan



Plastic has slowly became an integral part of all human requirements. Plastic carry bags, packaging material, bottles, cups, and various other items have slowly replaced everything made of other material due to the advantages of plastic.

Plastic is durable, easy to produce, lightweight, unbreakable, odorless, and chemical resistant.



Disadvantages

- Leaching of microplastics into soil and water bodies
- Accumulation by aquatic life

http://timesofindia.indiatimes.com/articleshow/84883086.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst

Kishore Class 4 · slide 21

Waste-to-Energy

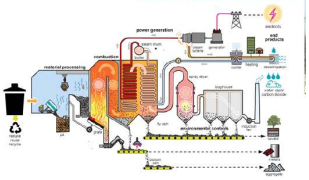
Vienna incineration plant



Waste-to-Energy facilities accept our solid waste and combust it at very high temperatures, producing heat that is used to convert water into steam. The steam is used to run turbines that generate electricity.

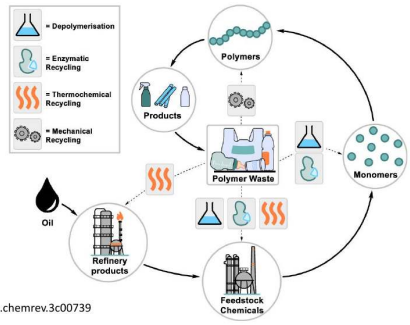
7 stages

1.Waste is dumped into a large pit.
2. dumps it in a combustion chamber.
3.The waste (fuel) is burned, releasing heat.
4.The heat turns water into steam in a boiler.
5.The high-pressure steam turns the blades of a turbine generator to produce electricity.
6.An air-pollution control system removes pollutants from the combustion gas before it is released through a smokestack.
7.Ash is collected from the boiler and the air pollution control system.



Kishore Class 4 · slide 22

Waste-to-Wealth Depolymerization



https://doi.org/10.1021/acs.chemrev.3c00739

Kishore Class 4 · slide 23

Fuel from plastics

- Pyrolysis
- Thermal decomposition
- Catalytic converters (Zeolites and cobalt)



Kishore Class 4 · slide 24

Landfill

- Today's landfills are very different from the old ones where people just dumped their garbage in an open area.
- Landfills are constructed and operated to strict environmental standards, including liners to protect groundwater.
- Within this hierarchy, landfilling waste is the lowest priority of the solid waste management options.



Kishore Class 4 · slide 25

Organic Chemicals in the Environment

Fate of polymers after use

- It has been estimated that plastics kill or injure tens of thousands of sea birds, seals, sea lions, and sea otters each year, and hundreds of whales, porpoises, bottlenose dolphins, and sea turtles
- A large portion of the reported fatalities are due to the entanglement of birds and mammals in the drift nets of commercial fishing boats

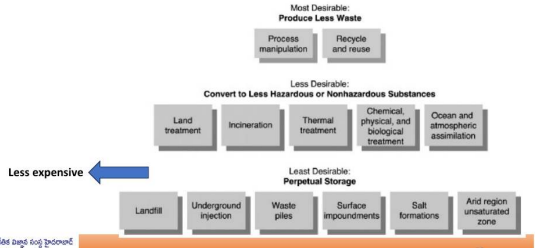
Possible solutions

- Development of biodegradable polymers
- Increase recyclability
- Reducing the usage of single use plastic

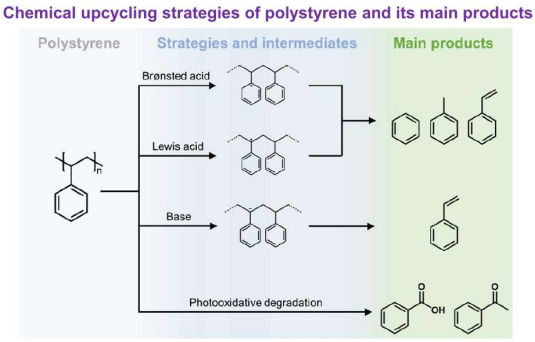
Kishore Class 4 · slide 27

Policy for Management and Disposal of Hazardous Waste

To minimize the amount produced,
(2) to convert the waste to a less hazardous or non hazardous form, and
(3) to isolate the waste in a secure, perpetual-storage site

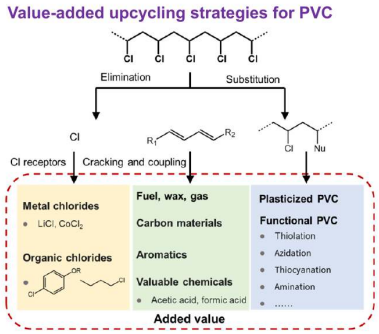


Kishore Class 4 · slide 29



Energy and Environment **2025**, *2* (1), 4. https://doi.org/10.53941/see.2025.100004 30

Kishore Class 4 · slide 30



Energy and Environment **2025**, *2* (1), 4. https://doi.org/10.53941/see.2025.100004 31

Kishore Class 4 · slide 31

Part C: Number sheet, traps and quick-fire drill

Number sheet

Quantity	Value
Haber-Bosch	200 atm, 450 °C, Fe/Mo; Nobel 1918 (Haber), 1931 (Bosch)
Photosynthesis ΔH	+470 kJ/mol
Free CO stretch	2143 cm^{-1}; metal carbonyls 1670-2057; $[Ti(CO)_6]^{2-}$ 1747
CO vs O_2 binding	free heme ~25,000×; Hb ~200× (slide on derivatives)
CO residence time / soil sink	2 weeks to 3 months / ~40%
SO_2 residence time	10-30 days
Air/fuel ratio	14.7 : 1
2-stroke unburned fuel; HC vs 4-stroke	25-30%; >12×
OH per NO_2; per HC in smog	2; 4
Ozone chain length (OH, NO_x, Cl)	~40; ~10,000; ~100,000
O_2 photolysis / O_3 absorption	<240 nm, 498 kJ/mol / 220-330 nm, max 250-260 nm
Ozone layer; 1 DU; hole	20-30 km; 2.7×10^{16} molecules/cm^2; ~220 DU
Pinatubo	June 1991, 25 Mt SO_2
Stockholm Convention	2001: 12 POPs; +8 in 2009; +1 in 2011
DDT	1874 made; 1939 insecticidal; Nobel 1948; EPA 1970; US ban 1973
PCBs	US production stopped 1977; $>10^6$× bioconcentration
Detergent share of lake phosphate	20-25%
Plastic	2 Mt (1950) → >450 Mt; waste ~400 Mt/yr; 18% recycled / 24% energy / 58% landfill; ~12,000 Mt by 2050

Trap list (answers may hinge on these)

- ⚠ CO + OH → CO_2 is placed in the stratosphere on the slide but happens mainly in the troposphere; follow the slide only if the option does not say troposphere.
- ⚠ CO affinity: free heme 25,000× vs Hb ~200×. Read what the question says (free heme group vs hemoglobin).
- ⚠ Ocean plastic: 8 Mt (Class 1-3) vs 1-2 Mt (Class 4). Decomposition: 20-500 y vs 15-1000+ y.
- ⚠ Rh reduces NO_x; it does not oxidise CO or HC (Pt/Pd do). Converter described as 'three-way' on slides and 'two-way' in the 2024 question.
- ⚠ Brown gas: NO_2. N_2O_4 is colourless. NO is colourless.
- ⚠ DU is per cm^2 (2.7×10^{16}), not per m^2 (options in 2024 differ only by area unit and exponent).
- ⚠ Dithionic = $H_2S_2O_6$; dithionous = $H_2S_2O_4$; thiosulfuric = $H_2S_2O_3$.
- ⚠ Bridging carbonyl has **lower** ν(CO) than terminal. Lowest ν among the options is the anionic $[Ti(CO)_6]^{2-}$ if present, otherwise $V(CO)_6$.
- ⚠ Dirty dozen contains PCBs, dioxins, furans (not DDE or lindane). Toxaphene is cotton/livestock; HCB is a fungicide.

Quick-fire drill (cover the right column)

Question	Answer
Pressure/temperature in Haber process?	200 atm, 450 °C, Fe/Mo
Which radical is the 'detergent of the atmosphere'?	OH•
Name the five spheres.	Atmosphere, hydrosphere, geosphere, biosphere, anthrosphere
Role of Rh in converter?	Reduces NO_x to N_2 (with H_2 from HC + H_2O)
Role of Pt/Pd?	Oxidise CO → CO_2 and HC → CO_2 + H_2O (air injected)
Ozone absorption maximum?	250-260 nm
Why very long cell for ozone?	A = εbC; ppm-level O_3 needs large b
Bond order of CO?	3
Bridging vs terminal ν(CO)?	Bridging lower
Coordination of oxy-Hb Fe(II)?	Six (4 porphyrin N, proximal His N, O_2)
Why does CO bind poorly to Fe in Hb versus free heme?	Steric clash with distal His; linear Fe-C-O impossible; weak angular bonding

Question	Answer
Which NO_x is brown?	NO_2
OH radicals per hydrocarbon in smog?	4 (one NO_2 gives 2)
Major eye irritant in photochemical smog?	PAN
Chemical name of $H_2S_2O_6$?	Dithionic acid
1 Dobson unit?	2.7×10^{16} molecules/cm^2
Which has the longest ozone-destroying chain: OH, NO, Cl?	Cl (~100,000); NO ~10,000; OH ~40
Pesticide for cotton/livestock in US, formula?	Toxaphene, $C_{10}H_{10}Cl_8$
PCBs stand for?	Polychlorinated biphenyls
TCE?	Trichloroethylene
Which POPs are unintentional?	Dioxins and furans
High K_{ow} means?	Hydrophobic: bioaccumulates, sorbs to sediment
Highest log K_{ow} in the table?	Aldrin (6.5); then DDT 6.2
Soap-making reaction?	Saponification of triglyceride with NaOH; NaCl salting out
Why replace polyphosphate builders?	Eutrophication; replaced by zeolites
Nylon and PET are formed by?	Step-growth (condensation) polymerisation
Polyacrylonitrile forms by?	Radical chain-growth polymerisation
Waste hierarchy top to bottom?	Source reduction, reuse/recycle, compost, volume reduction, landfill
Fate of all plastic made (Geyer)?	18% recycled, 24% incinerated, 58% landfill/environment